



Measuring the risk build up in Armenian financial system: A Growth-at-Risk approach

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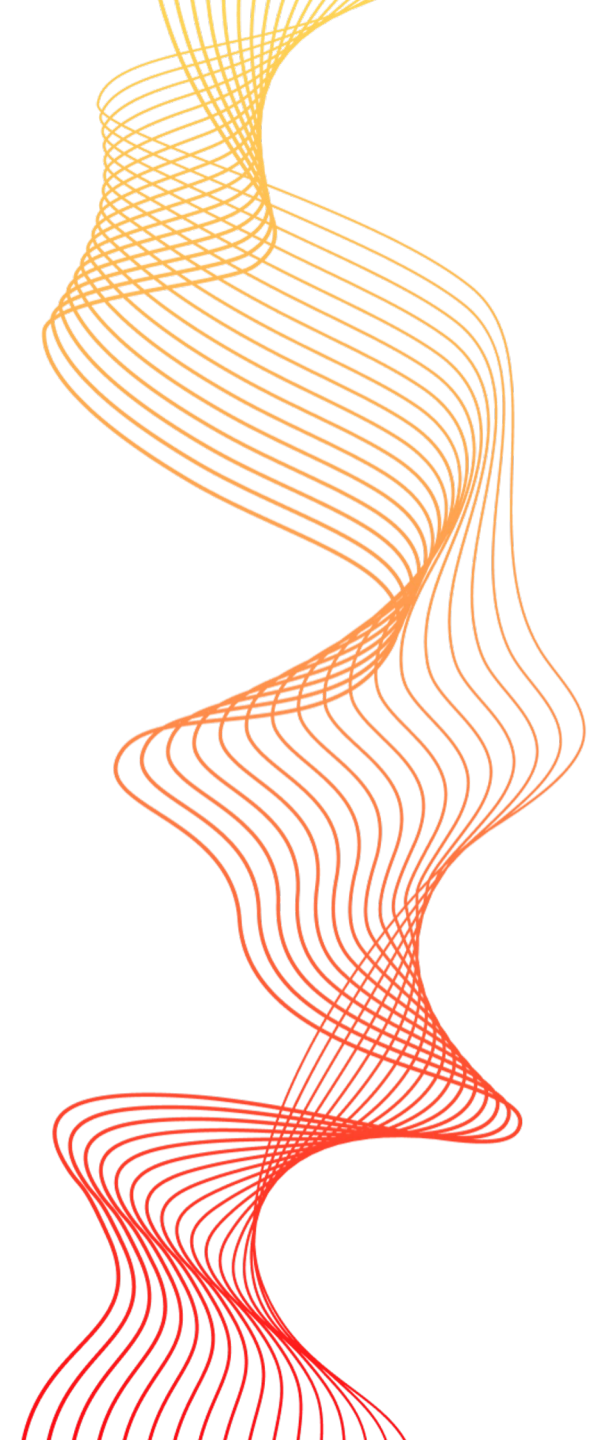
Important disclaimer: The views expressed in this paper are those of authors, and do not necessarily represent the views of the Central Bank of Armenia

What is it about?

Recent years have seen a rapid **operationalization of Macprudential policy (MP)** in many countries. But it still lacks reliable quantitative measurement framework and a criteria for policy rule (as in case of Monetary Policy).

The main problems of MP are that:

- Wide diversity of tools with **overlapping coverage**
- The impact is **difficult to quantify**
- Strong **non-linearities** in the effects of MP
- The absence of a clear and well-defined **target**



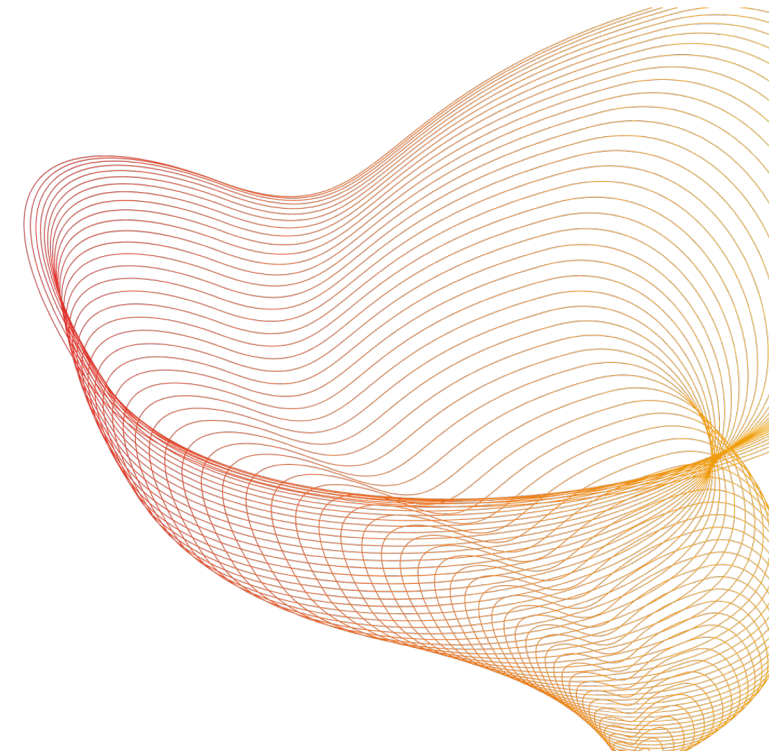
What is Growth-at-Risk?

In simple terms, Growth-at-Risk framework is used to model the impact of exogenous variables **not on the mean value** of dependent variable, **but on its entire distribution**.

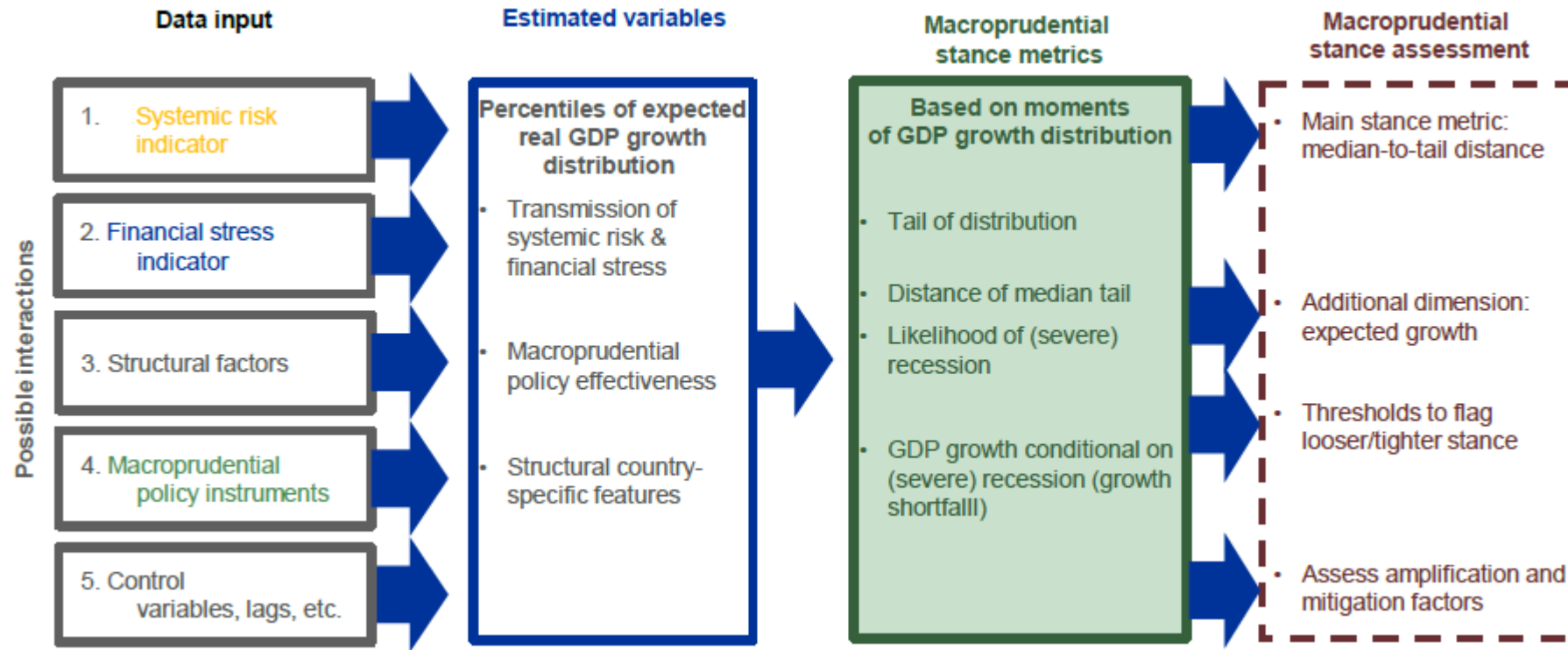
To do this, **it uses quantile regression approach** (Koenker & Basett, 1978).

The benefits of GaR framework:

- Measure **short-term costs vs long-term benefits** of Macroprudential policy tools,
 - The mean/median of distribution as indicator for **potential benefits**.
 - And the tail (usually 5th or 10th quantile) as a measure of **risks**
- Perform **policy simulations**,
- Design an intuitive **policy decision rule**



Growth-at-Risk Framework



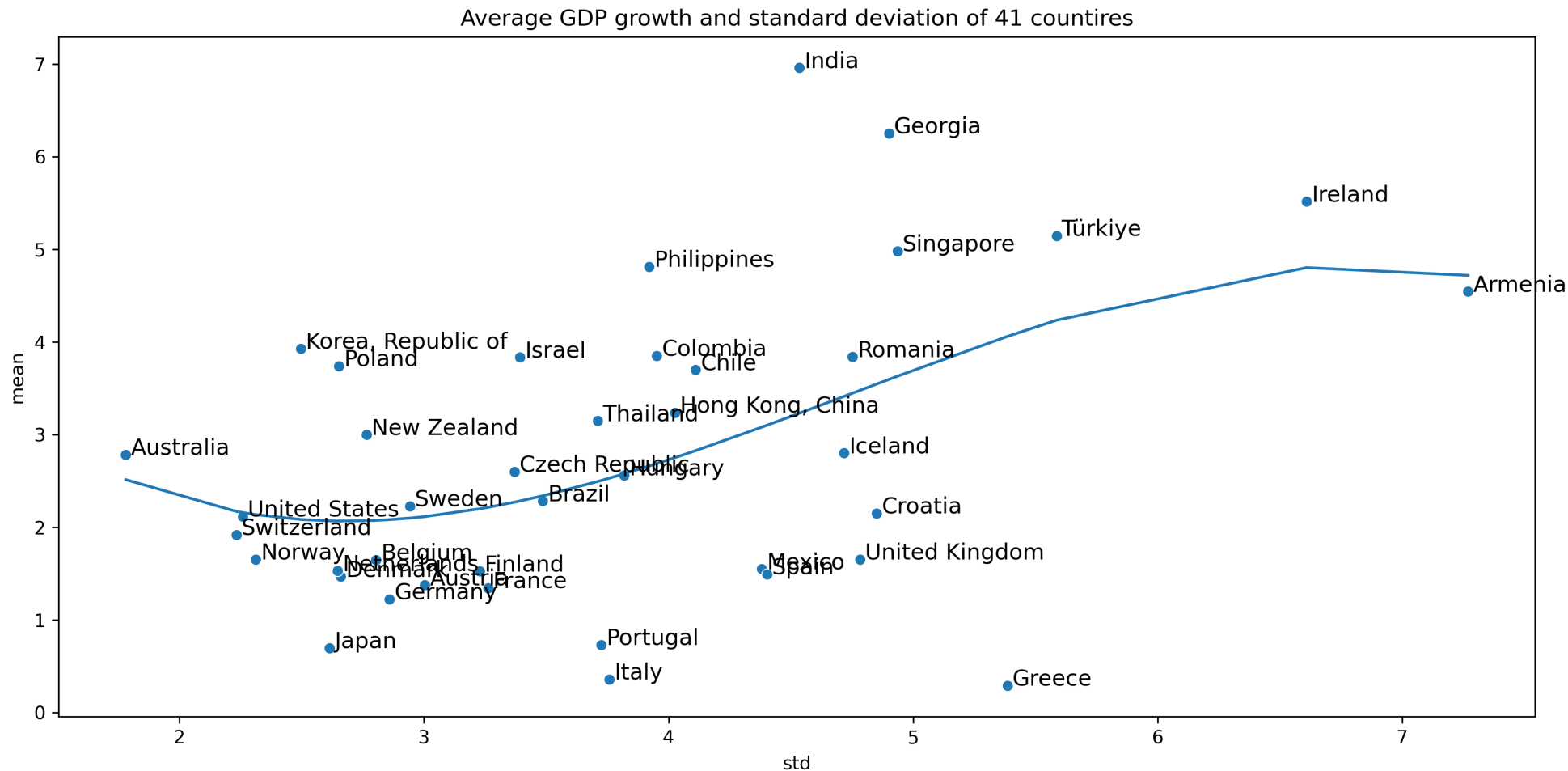
Source: ESRB

Data sources:

- IMF (FSI and IFS)
- World Bank
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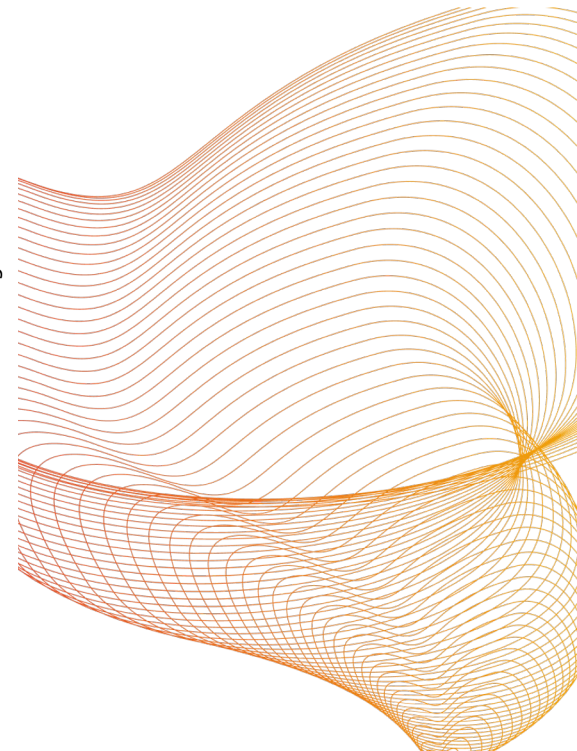
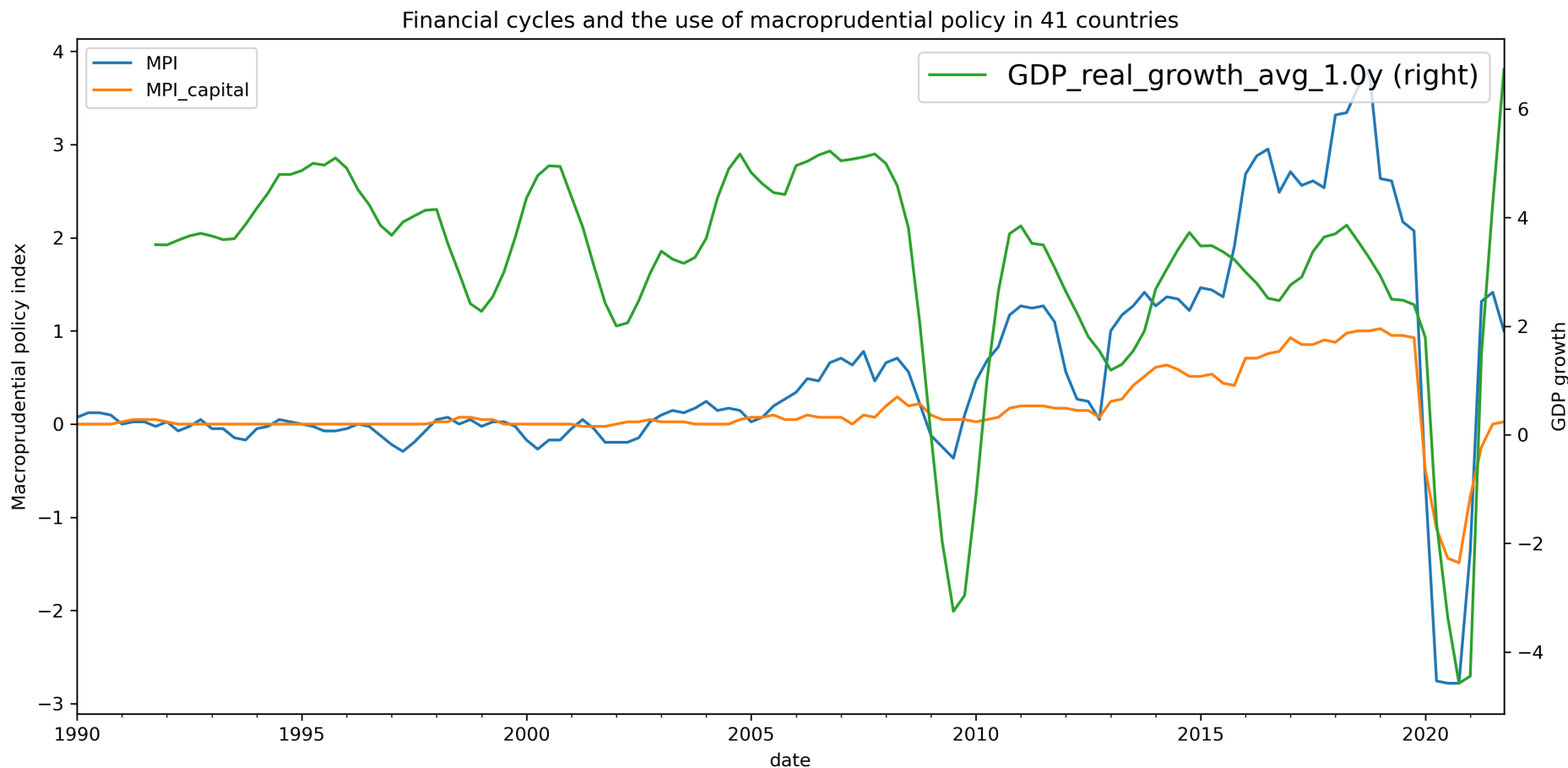
How to grow? Boom-busts vs steady growth paths

Going over the risk-reward relationship, we observe **diminishing returns** for countries with boom-bust growth fundamentals.



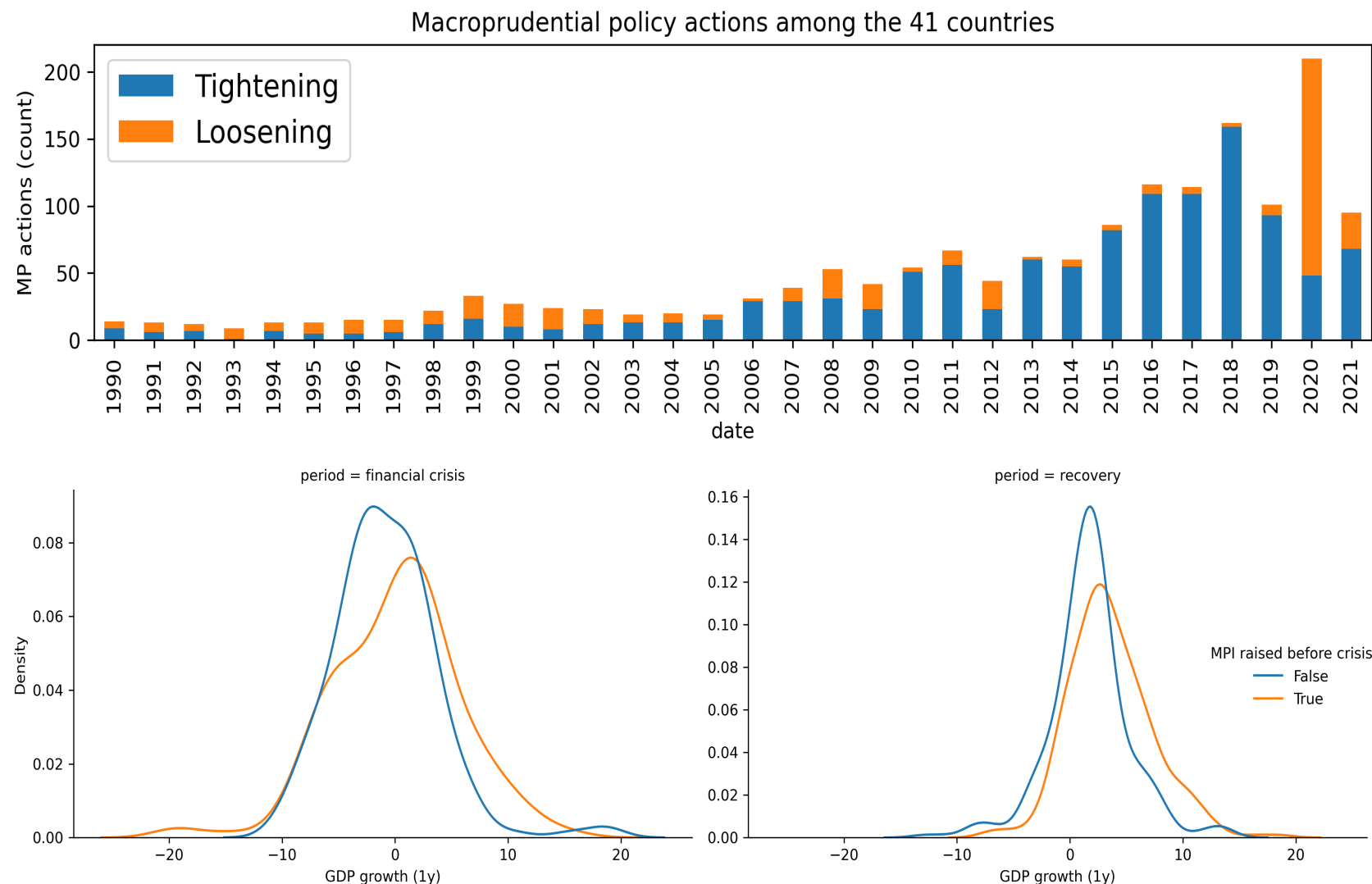
Use of Macroprudential policy

Over time MP used different tools. **Capital-based tools** were extensively used during COVID shock in 2020-2021.



Use of Macroprudential policy

Over time the policymakers **increased the usage** of MP tools. Mainly it has been **restrictive**, except for crisis periods (2008, 2020 etc.)



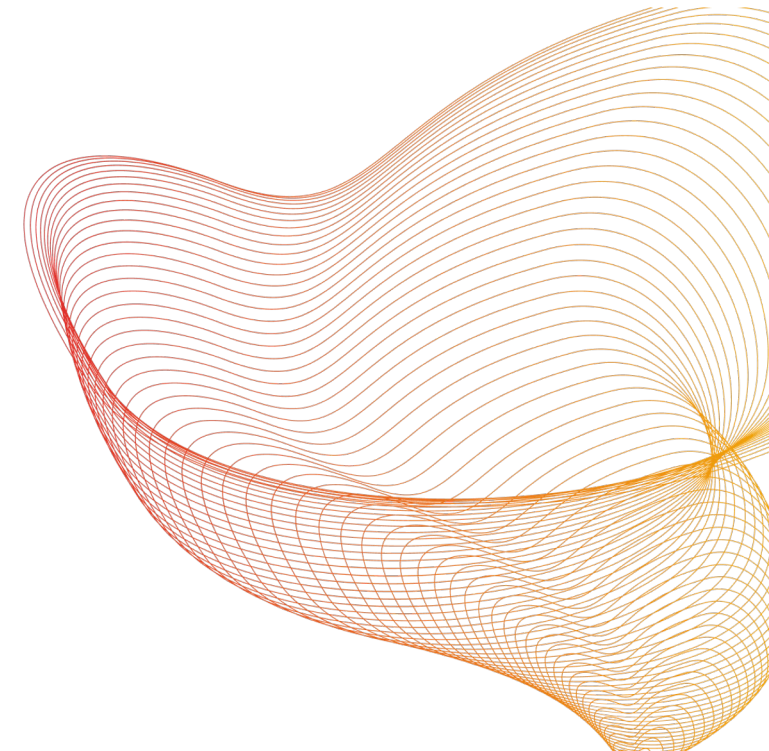
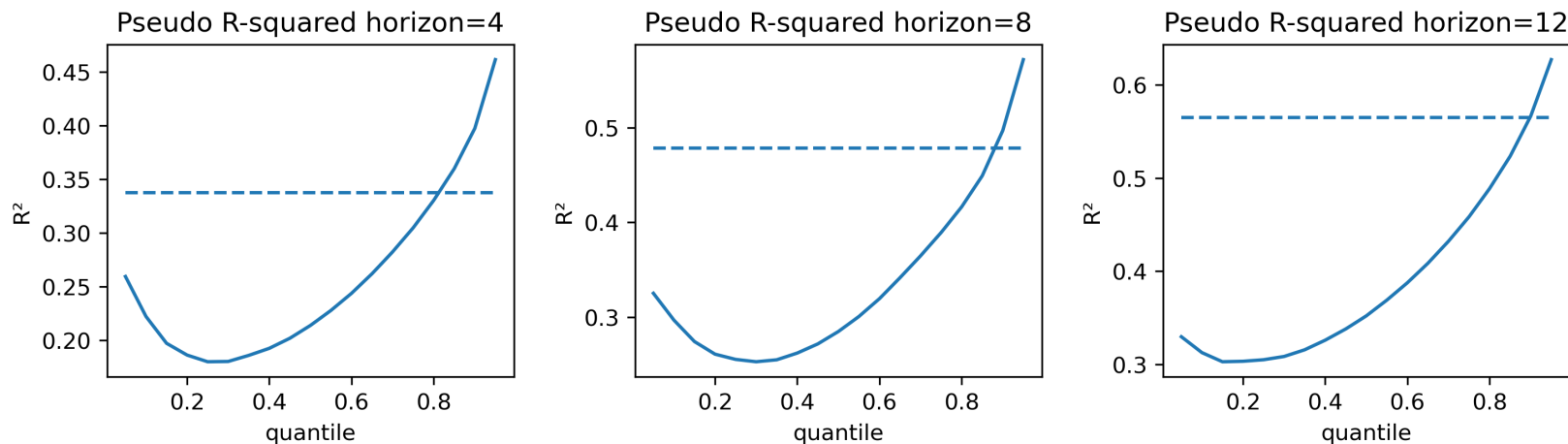
Panel model

The main benefit of **quantile regression** is that it allows us to model the non-linear relationships, that can affect the GDP distribution.

$$y_{i,t+h} = \alpha_{i,\tau} + X_{i,t}\beta_{\tau} + v_{\tau,t+h}$$

$$v_{\tau,t+h} = \sum_{i=1} \sum_{t=1} \rho_{\tau} |y_{i,t+h} - y_{i,t+h}^*|$$

The pseudo R2 for our model is around



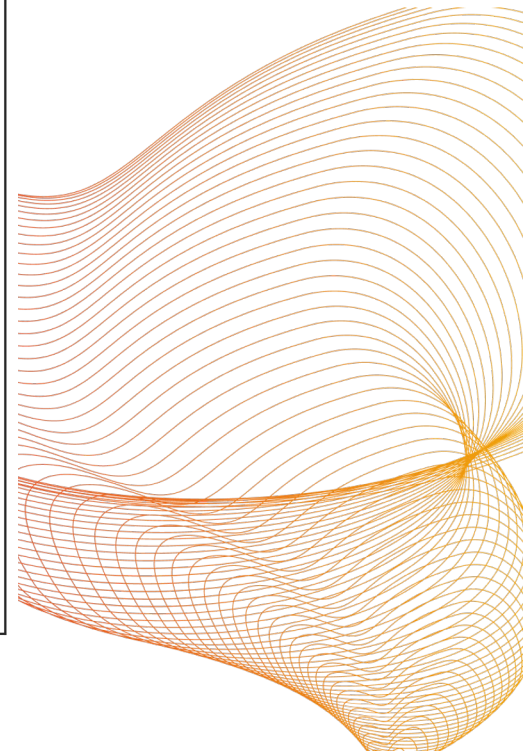
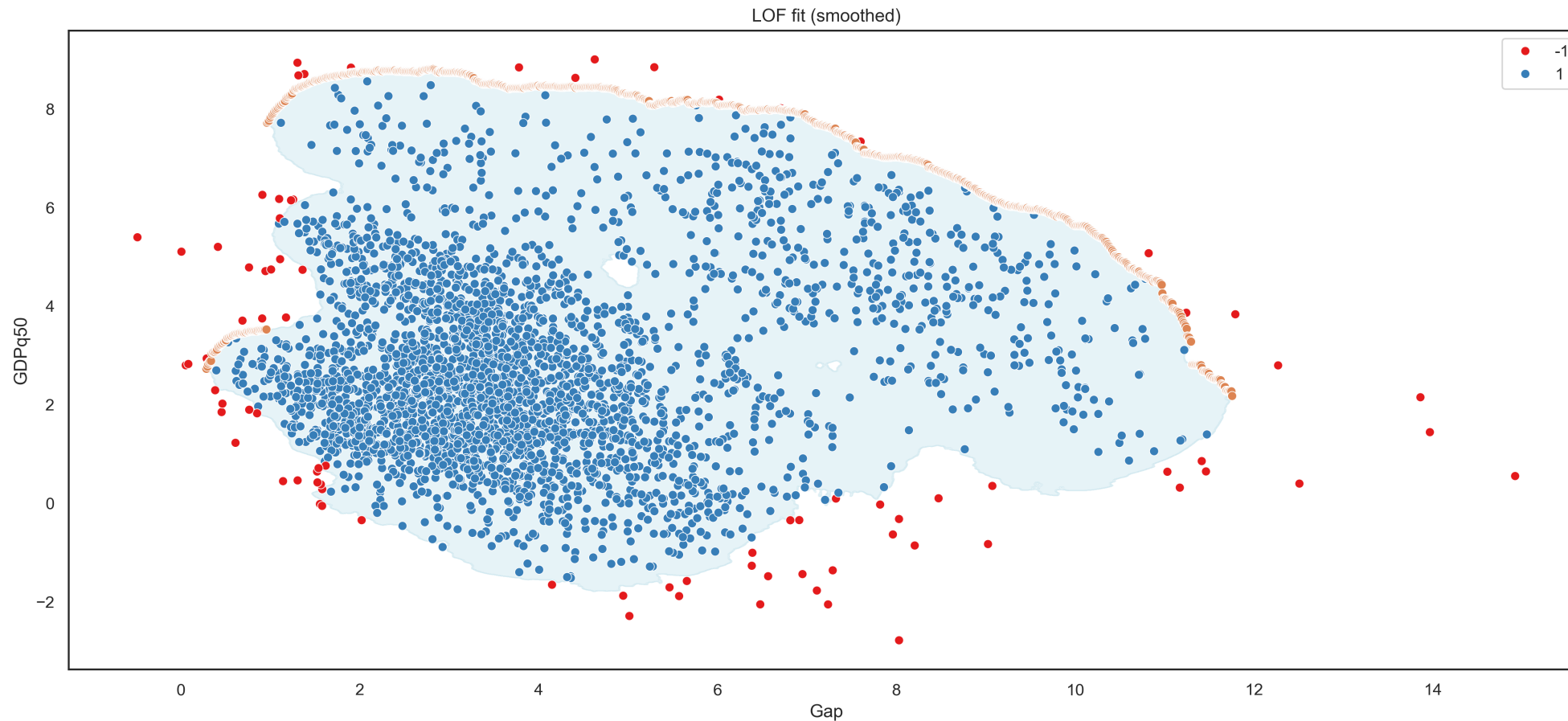
Armenia risk-return vs world

According to panel estimations, Armenia's growth pattern is generally in **boom-bust territory**.



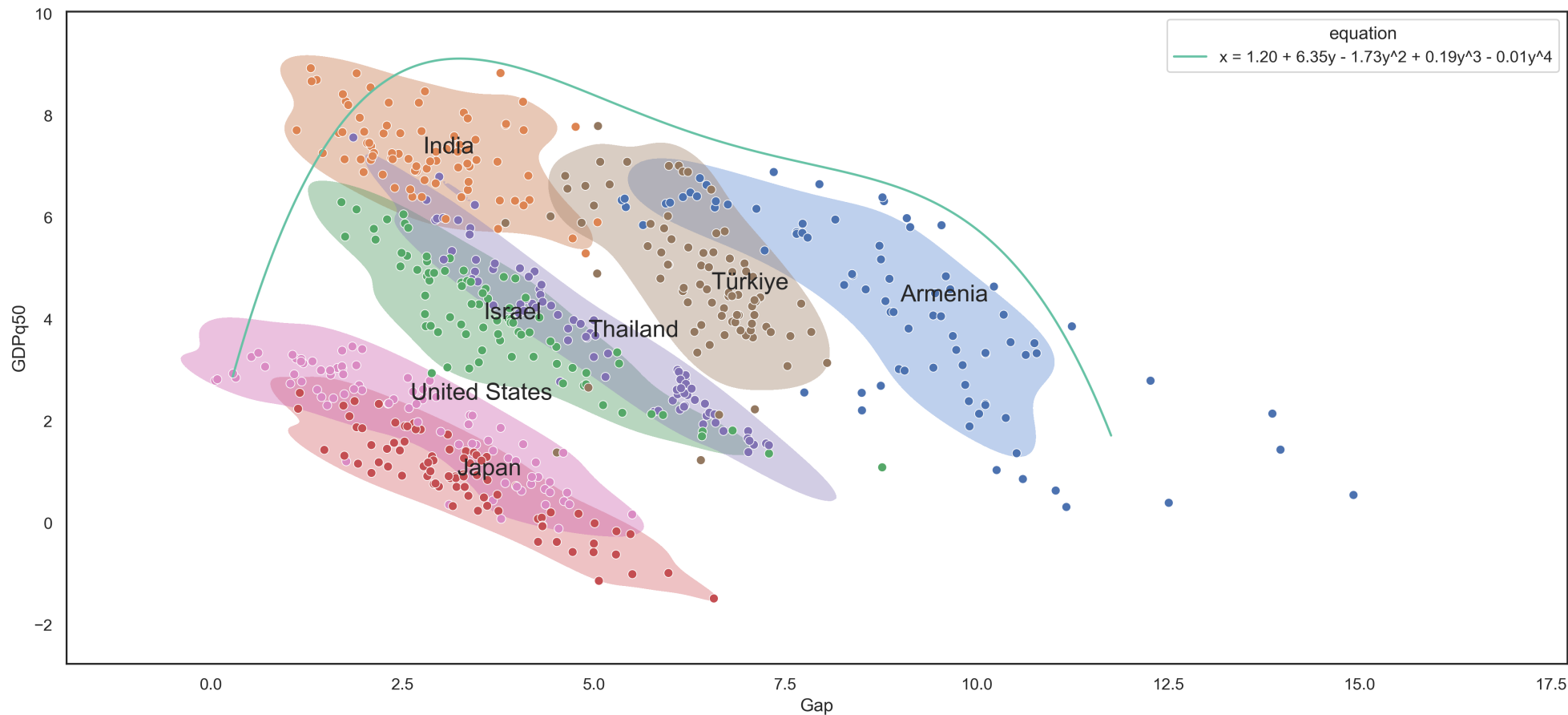
Frontier for best risk-return cases

Using LOF method, we estimate the frontier for the **best risk-return** performing countries



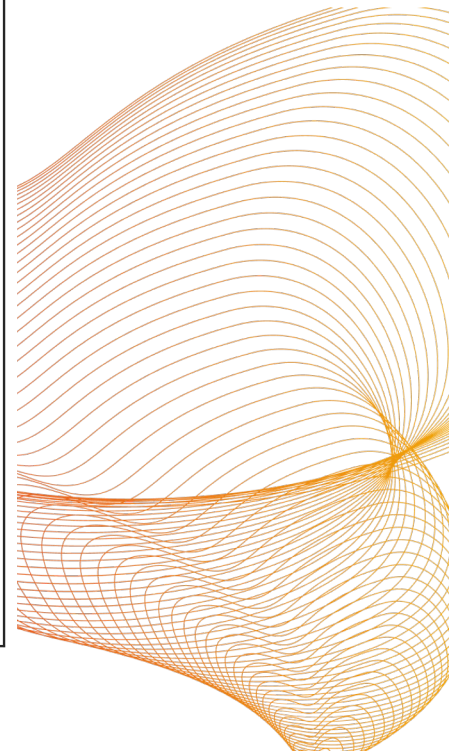
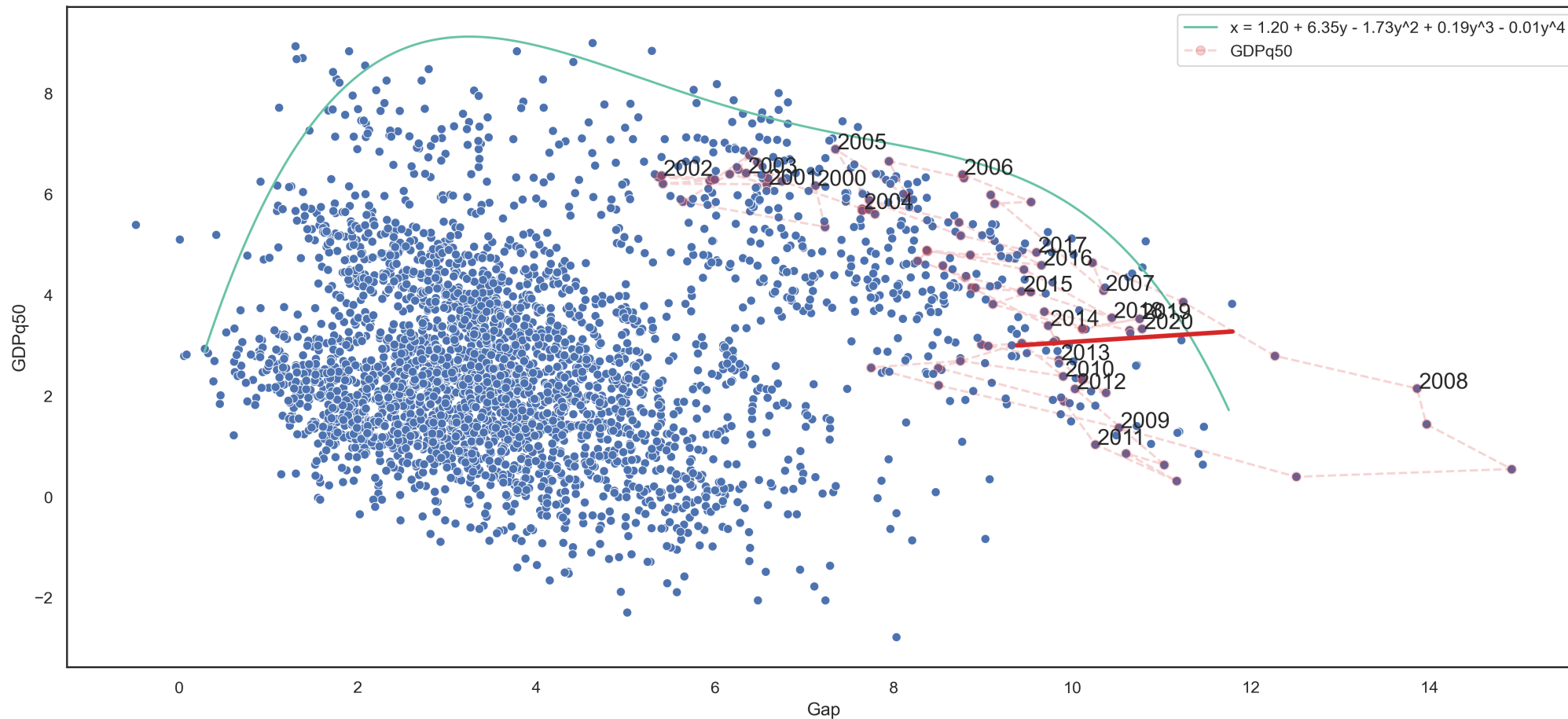
Country cases that shape the boundary

According to the estimations, beyond the 4-5% GDP shortfall, additional **risk-taking is suboptimal**. Countries should aim to manage their risks so that it is less than this threshold.

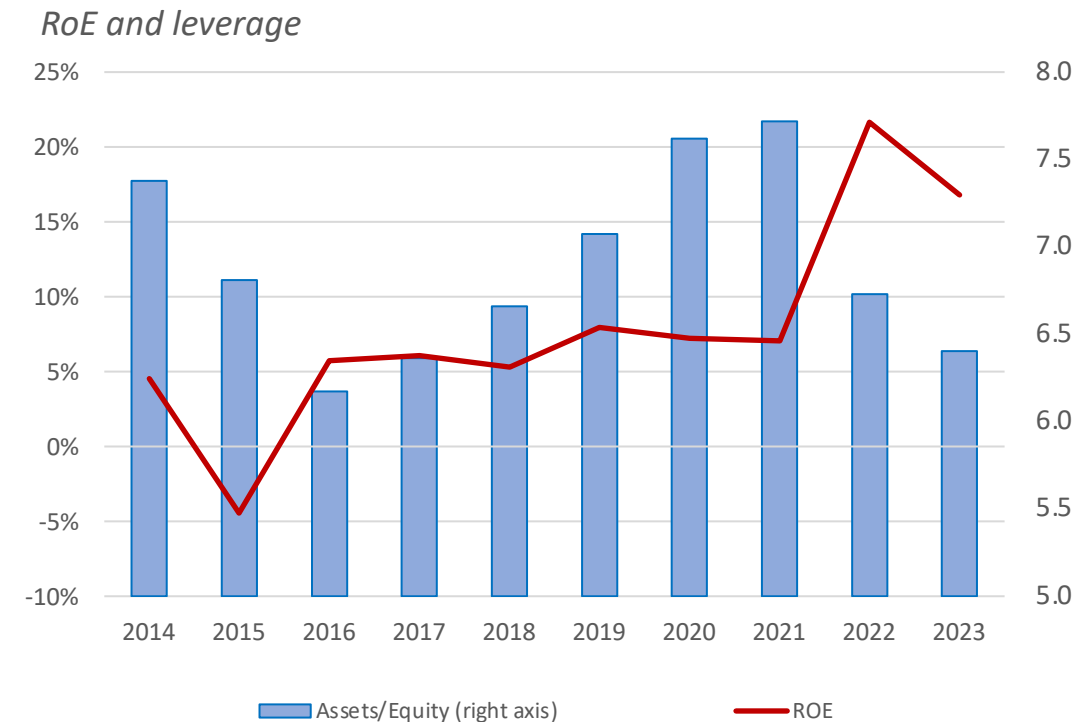
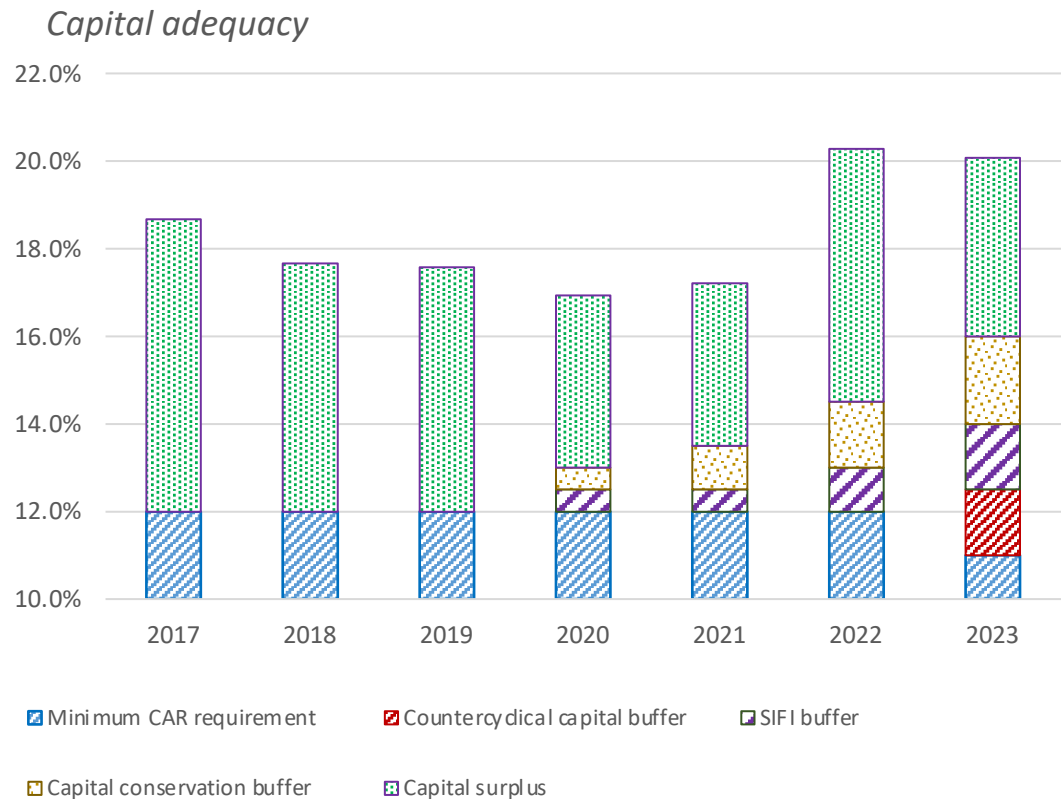


Possibility curve and the frontier

The **possibility curve** is bundle of the risk-return values that are **achievable** through the tightening and loosening of MP



Risk absorption capacity remains high

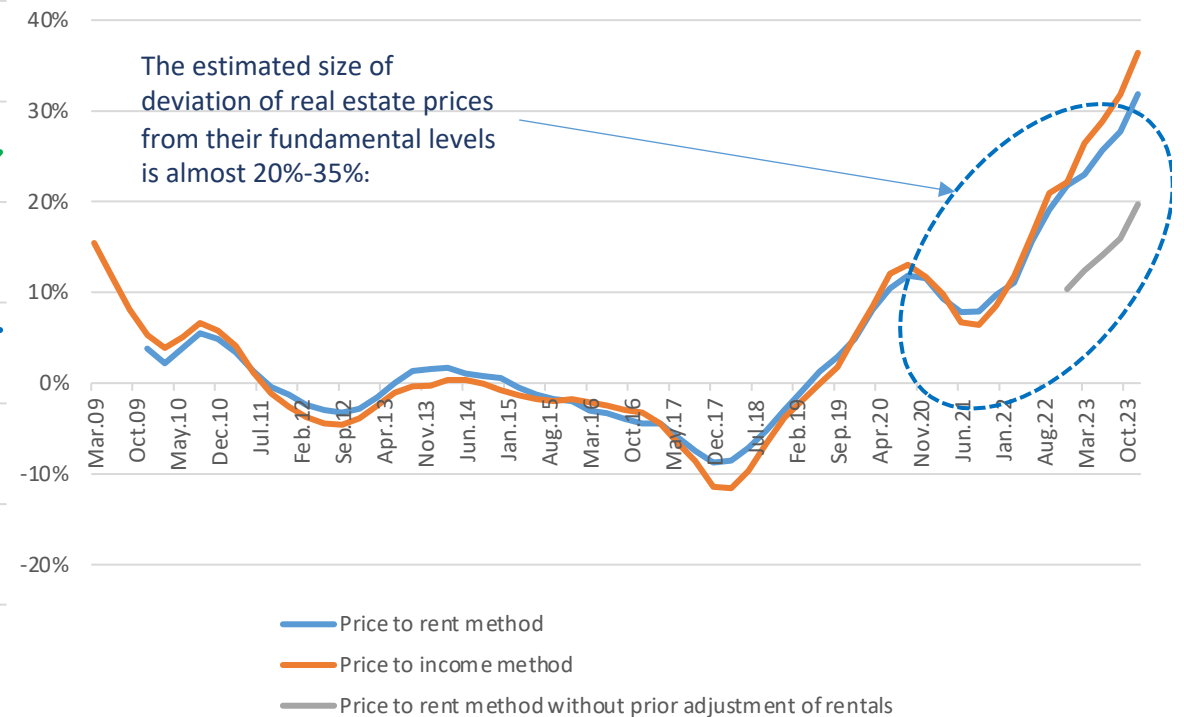


Real estate prices have grown. There are possible overvaluation risks

*The growth rate of average real estate price**



*The estimated size of deviation of real estate prices from their fundamental levels ***

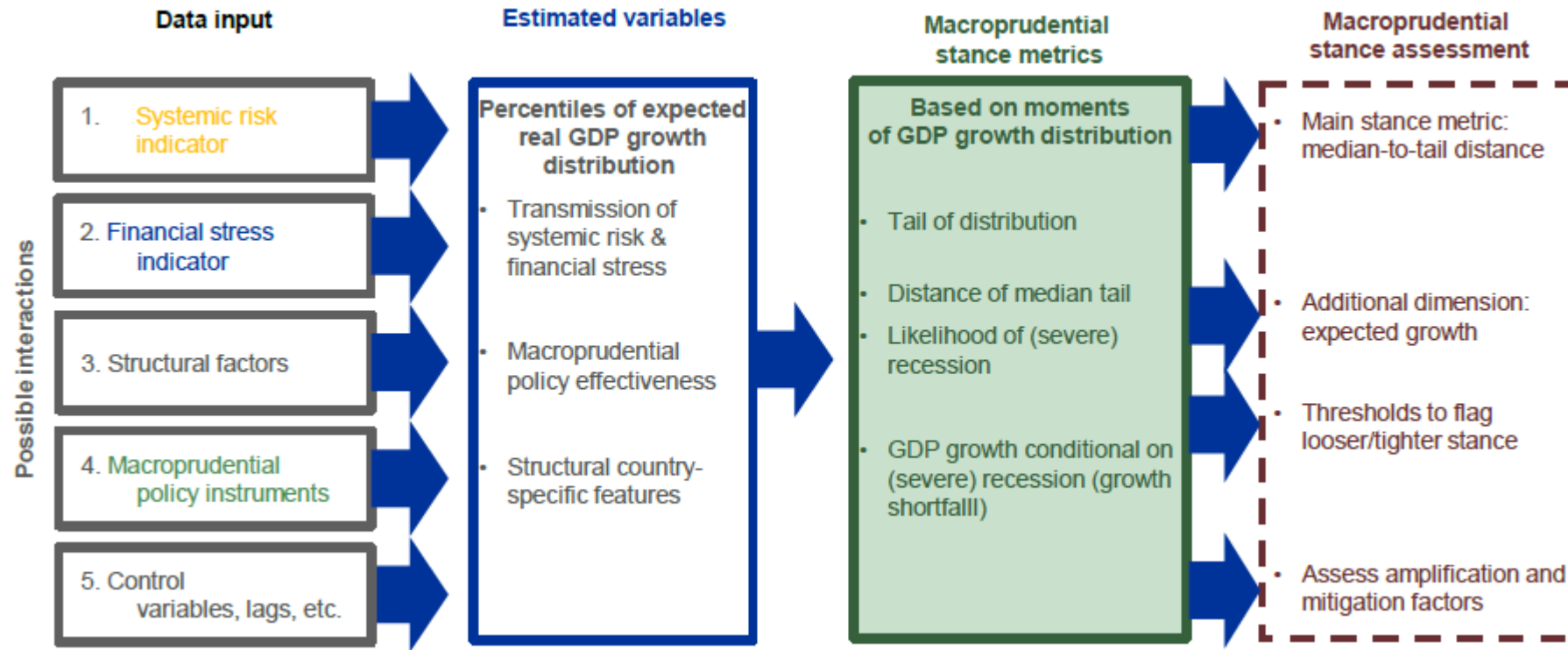


* For 1 sq. meter of multi-flat apartment building in Yerevan

Source` Real estate cadastre RA, CB calculations

**It is worth to mention, that this estimate is only a warning sign for possible overvaluation risks in real estate market and should not be interpreted as a forecast of real estate prices

Growth-at-Risk Framework



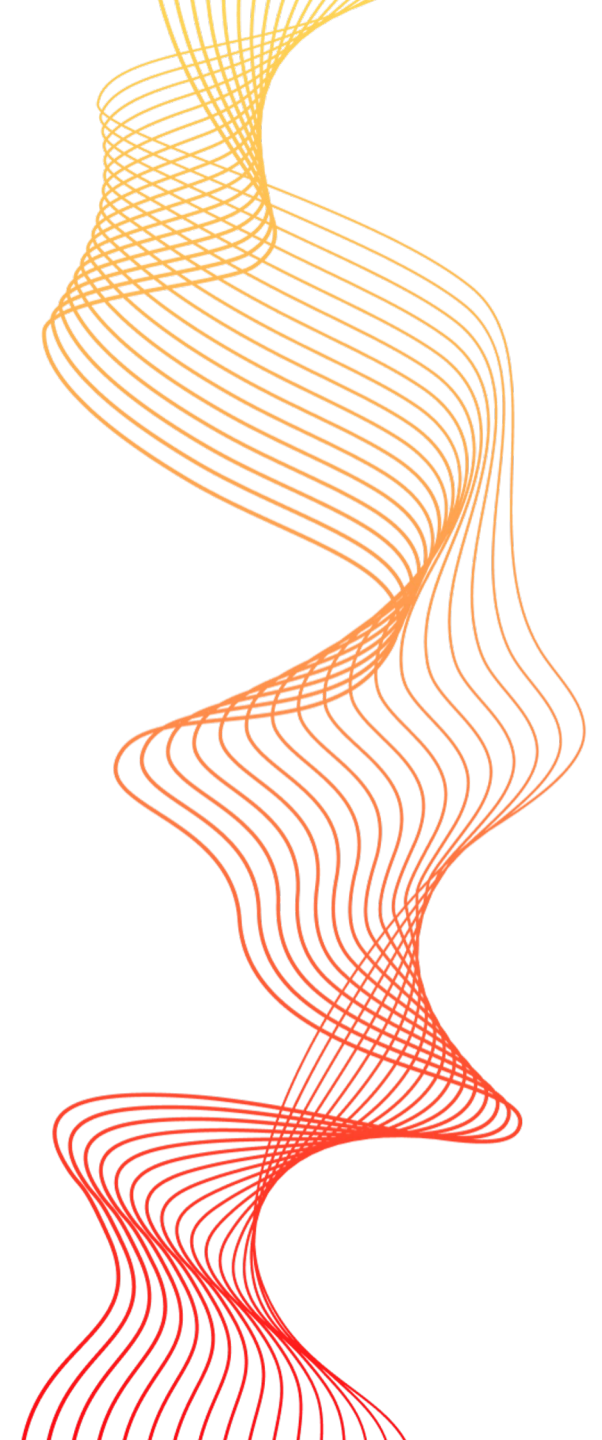
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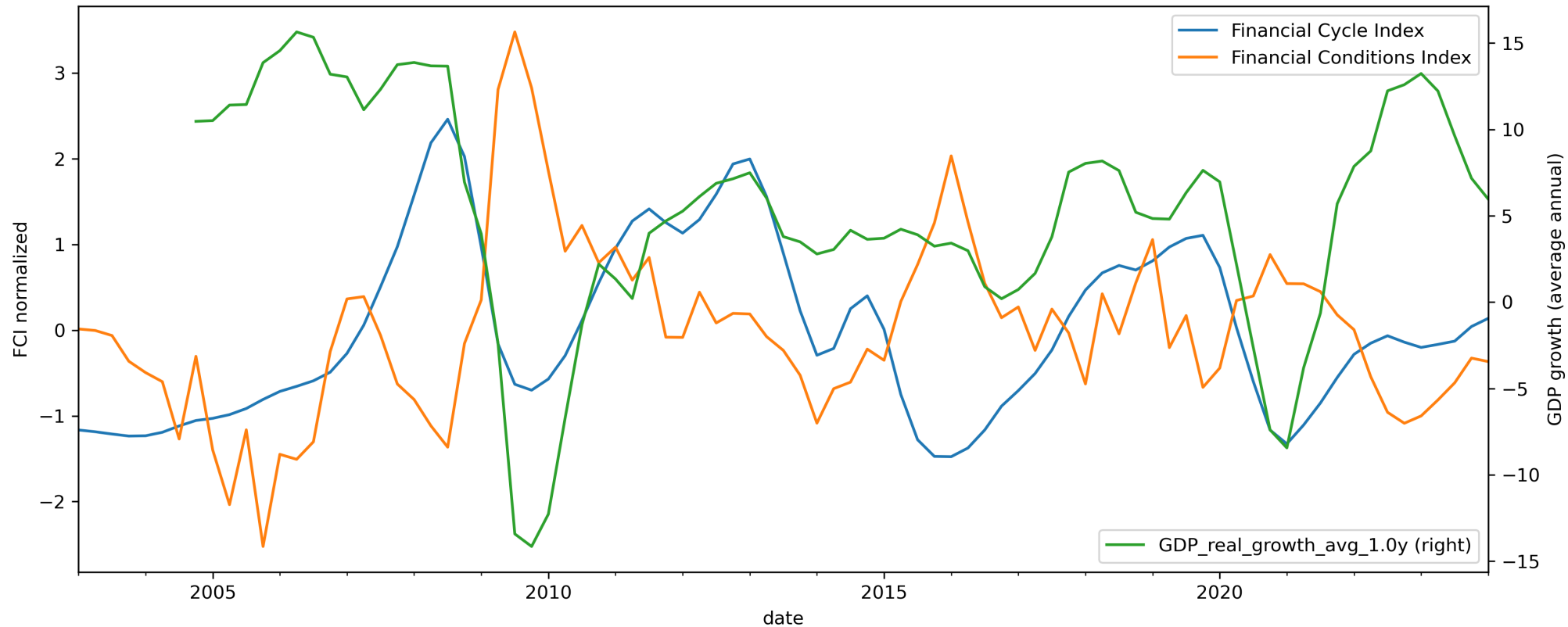
Key indicators

1. **Financial vulnerability** – captures cyclical system risk arising from credit and external imbalances. It has good early warning properties (Lang et al., 2019).
2. **Financial stress** – periods in which risks have materialized, reflected in stress episodes with larger asset price volatility, interest rate spreads, and deteriorated risk sentiment. Incorporates variables from financial markets, including interest rates with different maturities (Koop and Korobilis, 2014).
3. **Macroprudential policy index** – a dummy variable, where a value of -1 indicates loosening, $+1$ tightening and no actions are defined as 0. To account for the persistence in MP tools, a 4 quarter rolling sum is used (Galan, 2020).
4. **Structural variables** – includes a range of variables from financial sector, such as indicators for risk absorption capacity (capital, liquidity, profitability ratios), as well as risk-related variables (NPL ratio etc.).
5. **Control variables** – variables from macroeconomy



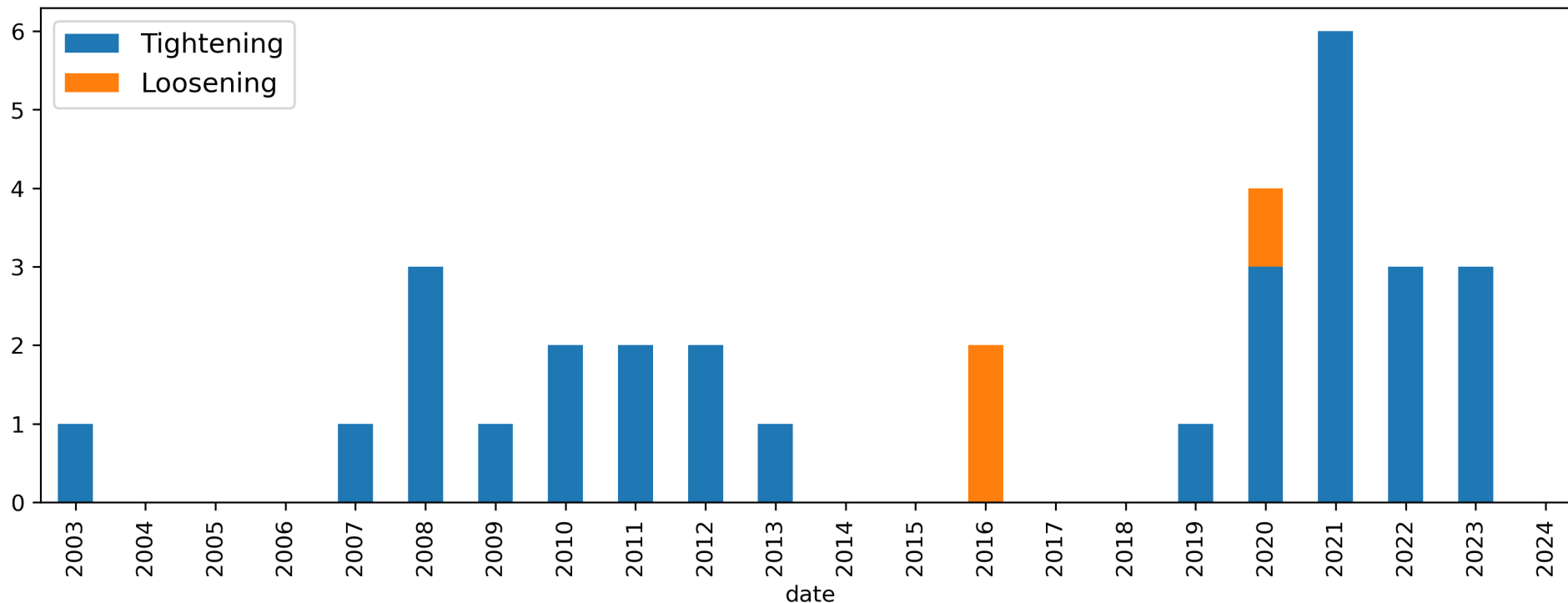
Financial vulnerabilities, stress and real GDP growth

The **Financial cycle index** tracks the risk build-ups in our financial system and is used as a forward looking indicator. The **Financial conditions index** is meant to show the current tension level in the financial markets.



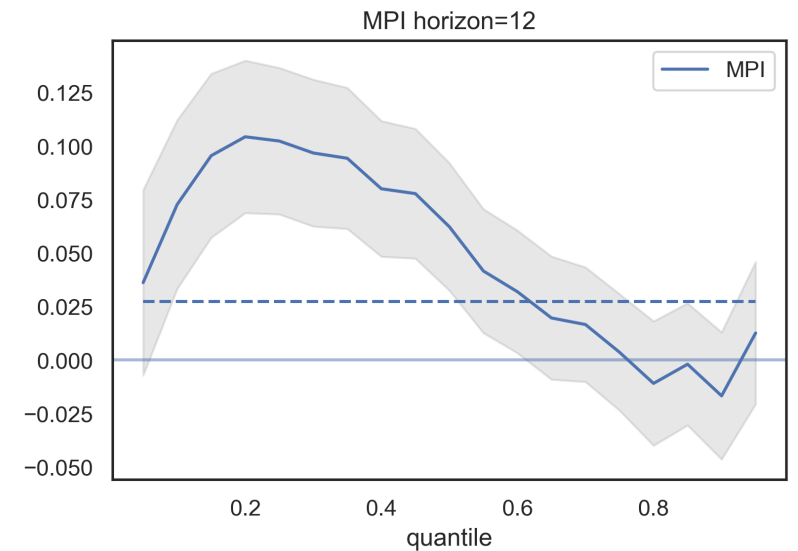
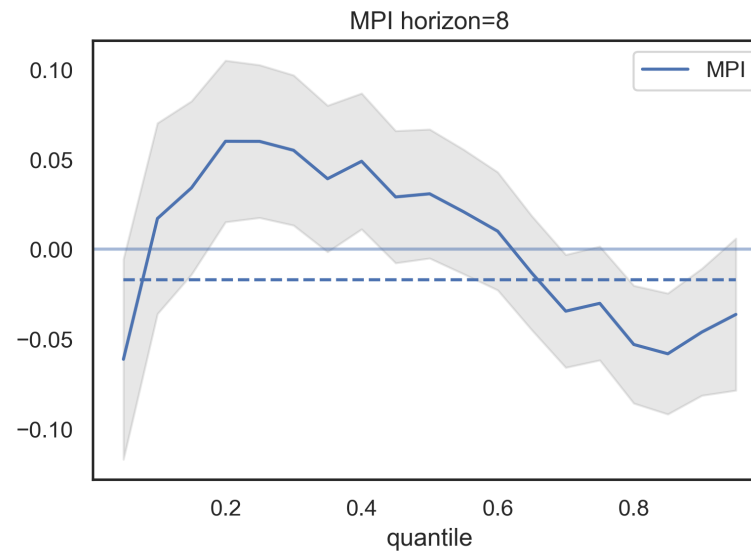
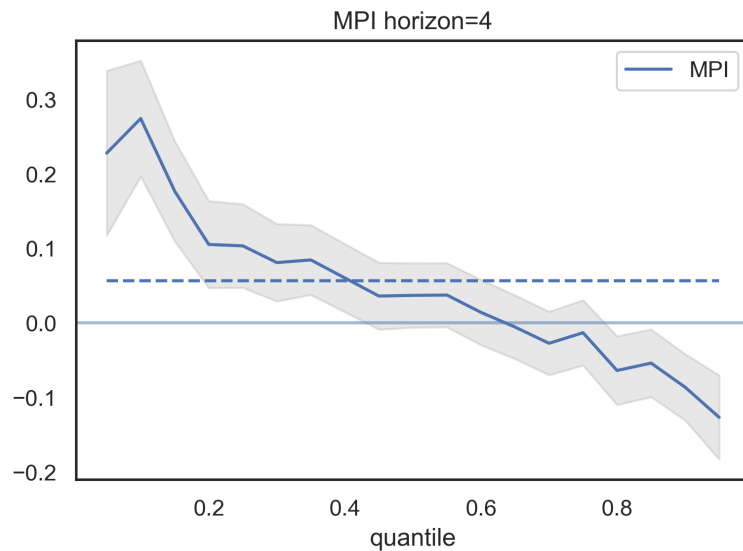
Macprudential policy usage in Armenia

The MP has been increasingly used **after the Great Recession** of 2008, as a result the MP history in Armenia is short. To improve the robustness of the results, we generalize the MP coefficient using the **panel estimates**.



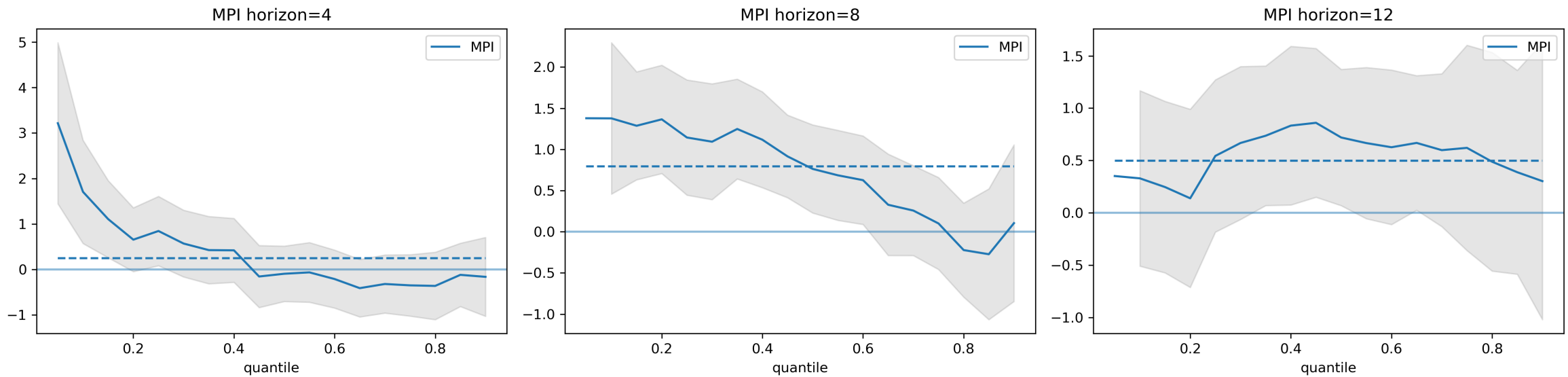
Panel estimation results: The effect of MPI

In the short-run the tightening of MP **mitigates the systemic risks** (the left tail shrinkage), and has an **insignificant impact on the median**. A similar pattern can be observed for the long run effects.



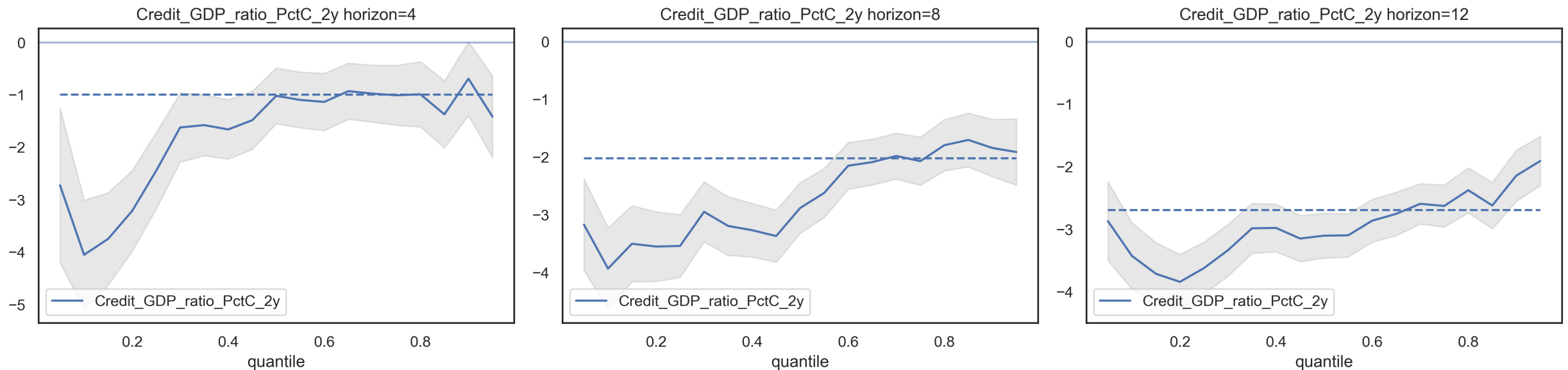
Local estimation results: The effect of MPI

The **short-run** local model results are **in line with panel** estimates. The coefficients for the **long-run** model indicate **positive** effects across the entire distribution.



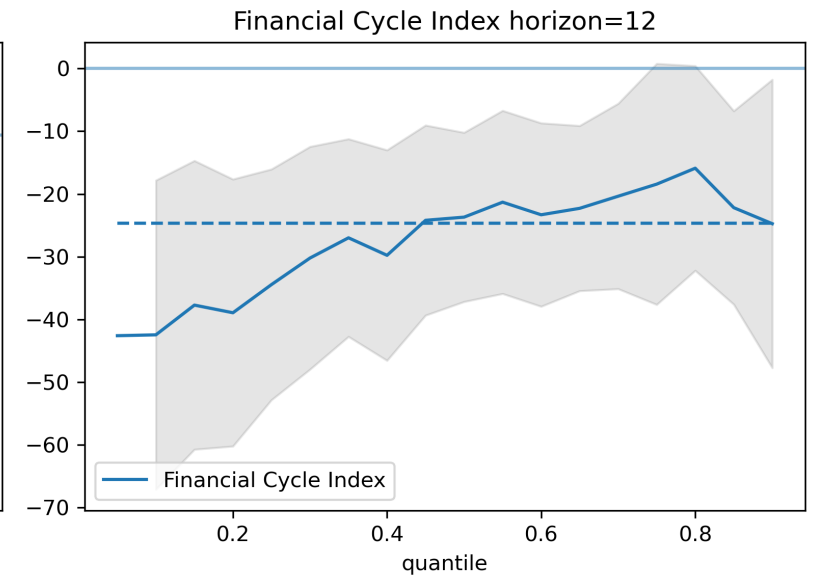
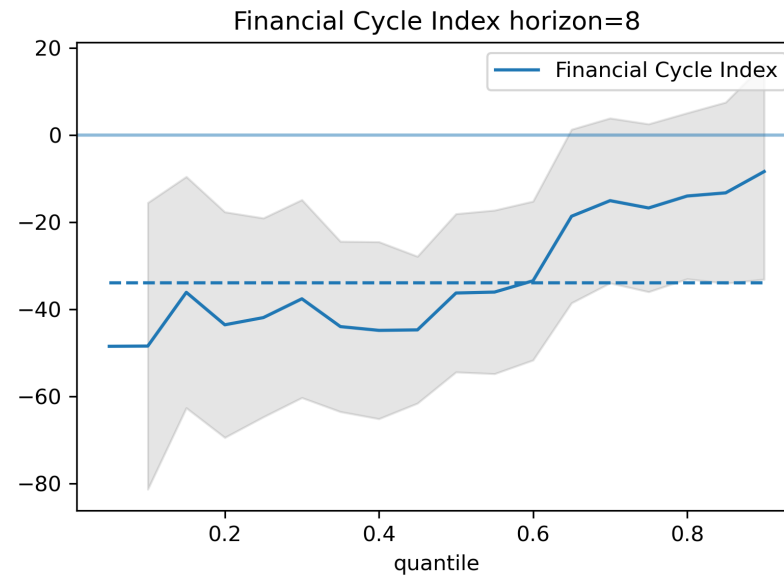
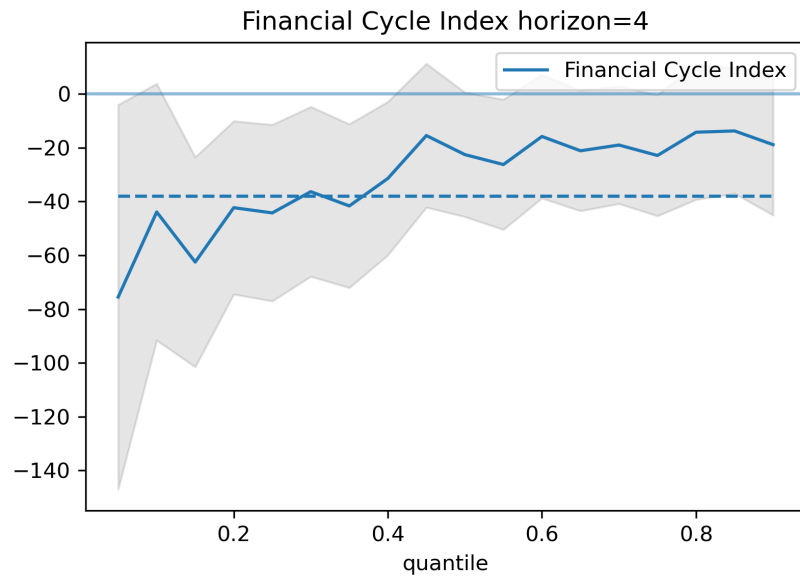
Panel estimation results: the effect of financial vulnerabilities

A positive shock to financial intermediation (excessive credit growth) causes the **risks to rise** in both short- and long-term, to simultaneously **deteriorate the average GDP growth** prospects in the long-run.



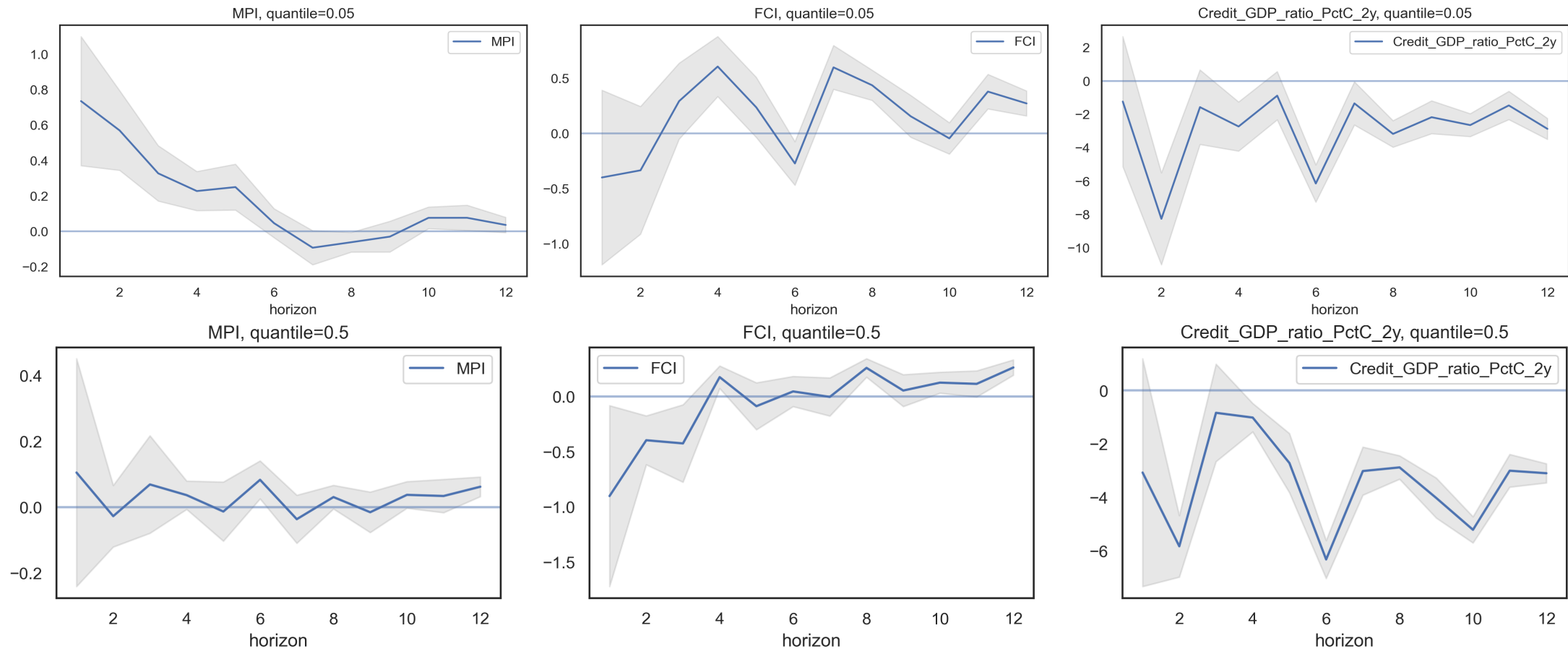
Local estimation results: the effect of financial vulnerabilities

The **deteriorate** the average GDP growth as a result of excessive credit growth continues to **persist** in the long-run.



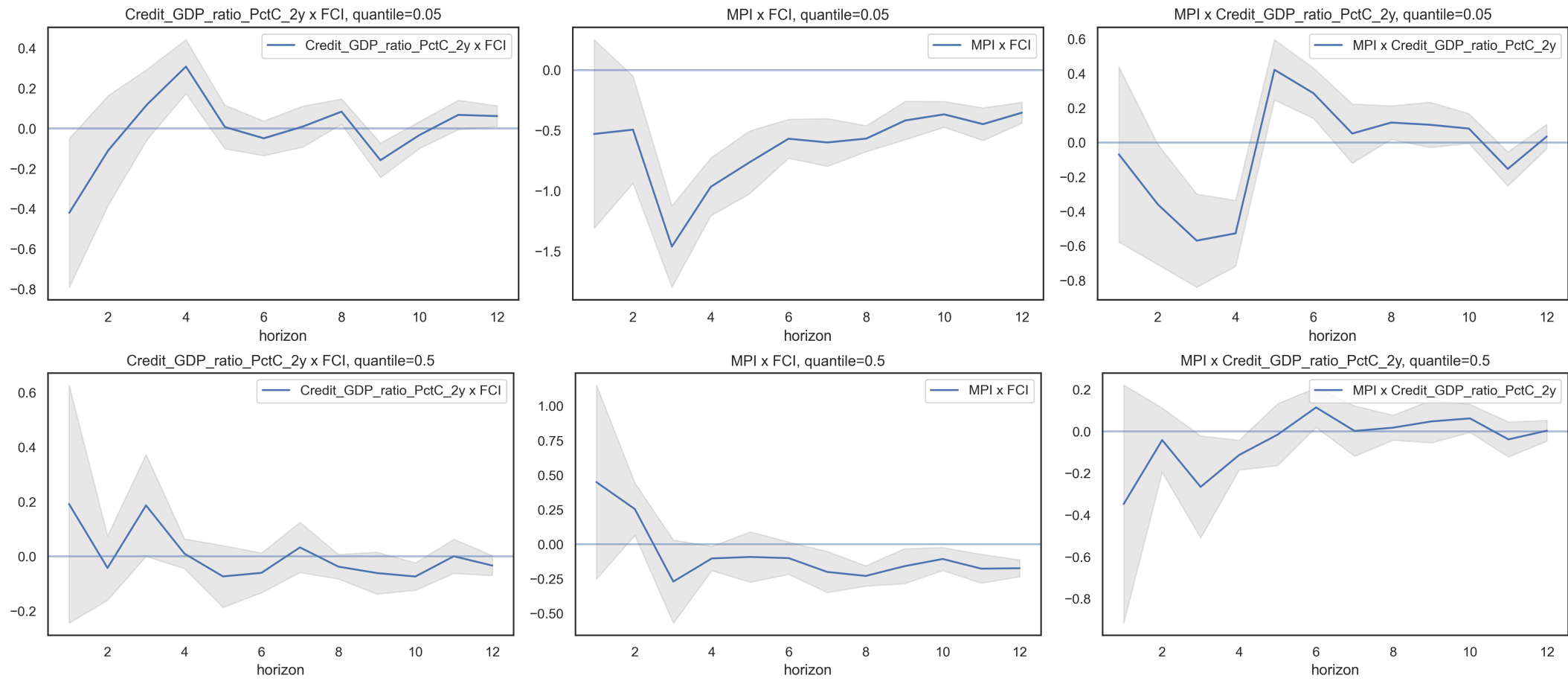
Panel estimation results: effects by horizons

The usage of MP **heavily affects the risky tail** of GDP growth distribution within 1.5 years, while having **virtually no impact on average GDP growth**.



Panel estimation results: interaction effects

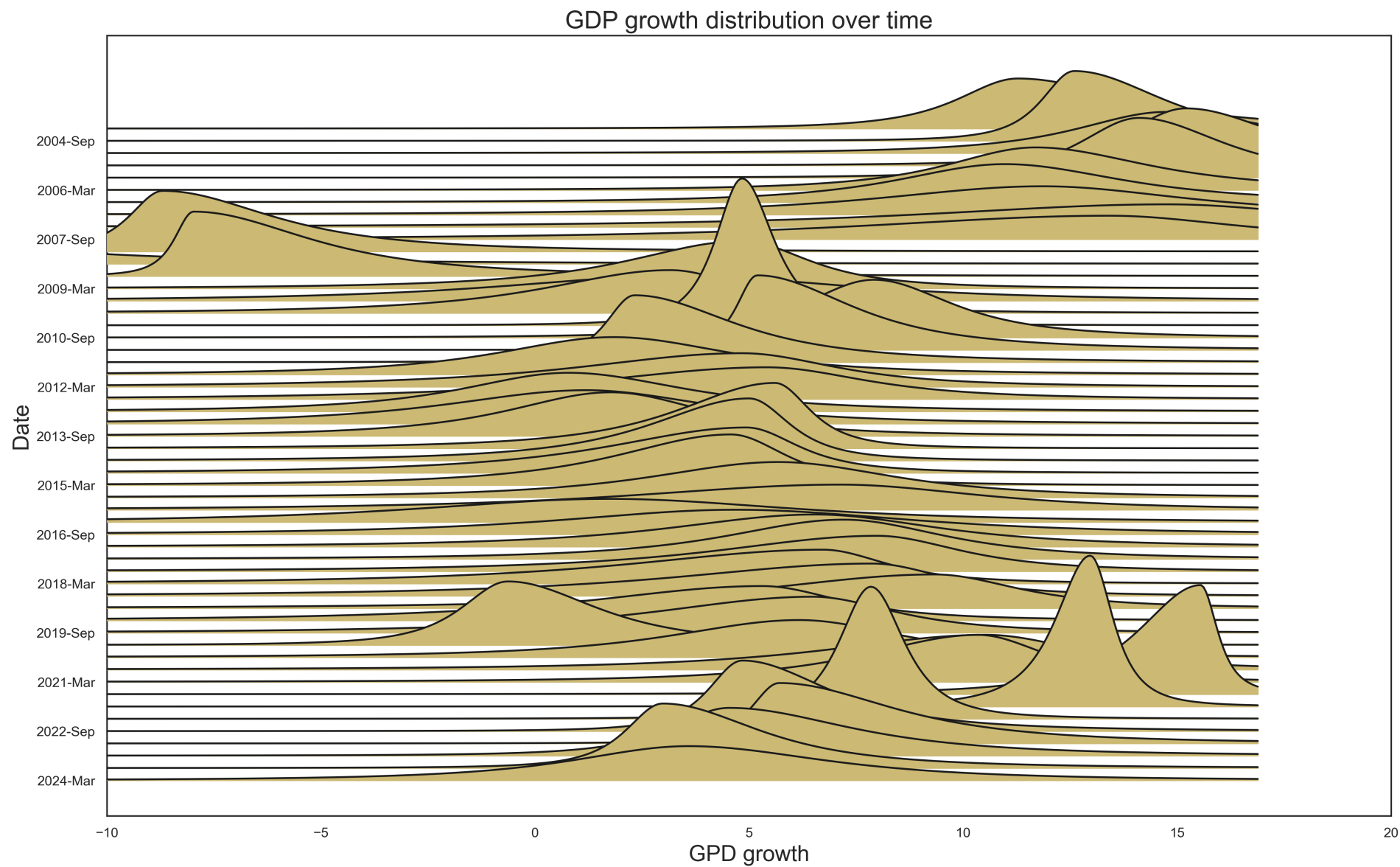
According to the estimates the **loosening of MP when the financial conditions are tight** (stress is high), will have a **positive effect** on the tail reducing the risks up to 0.5% in the long-run.



Historical quantile fit

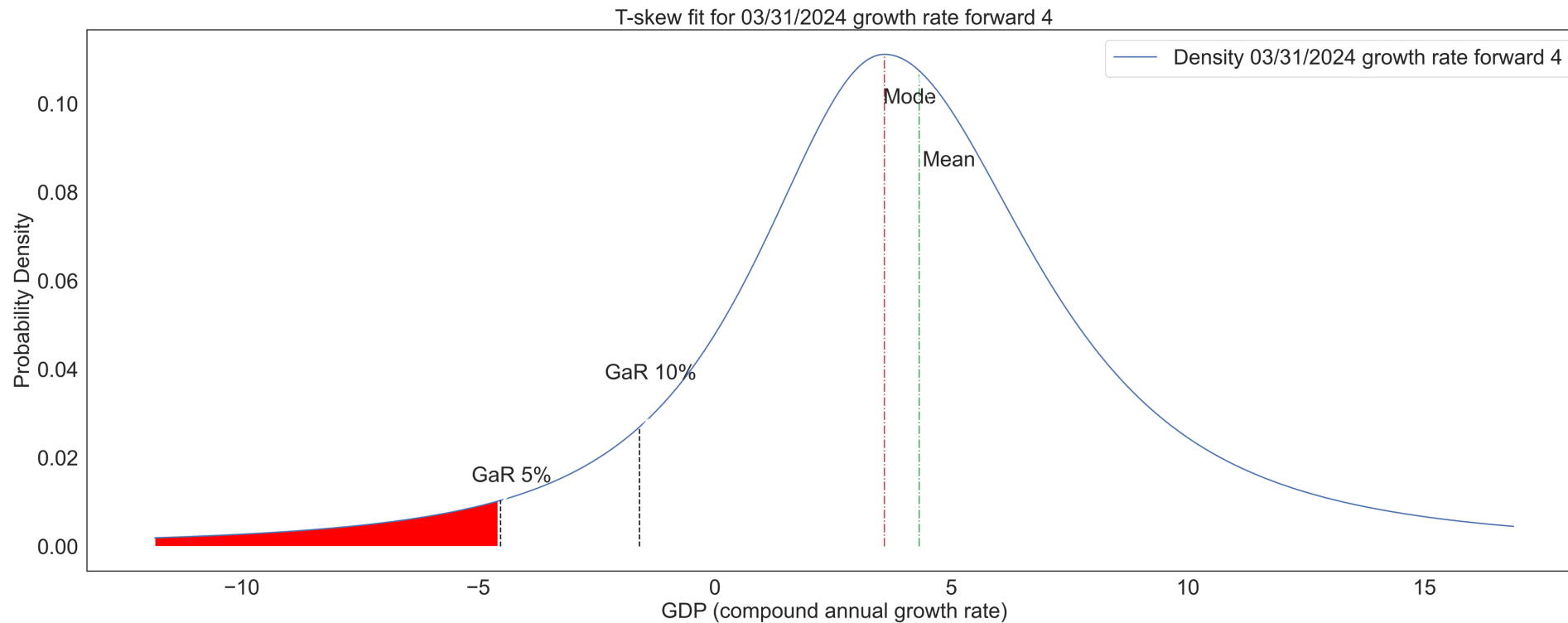


Historical quantile fit



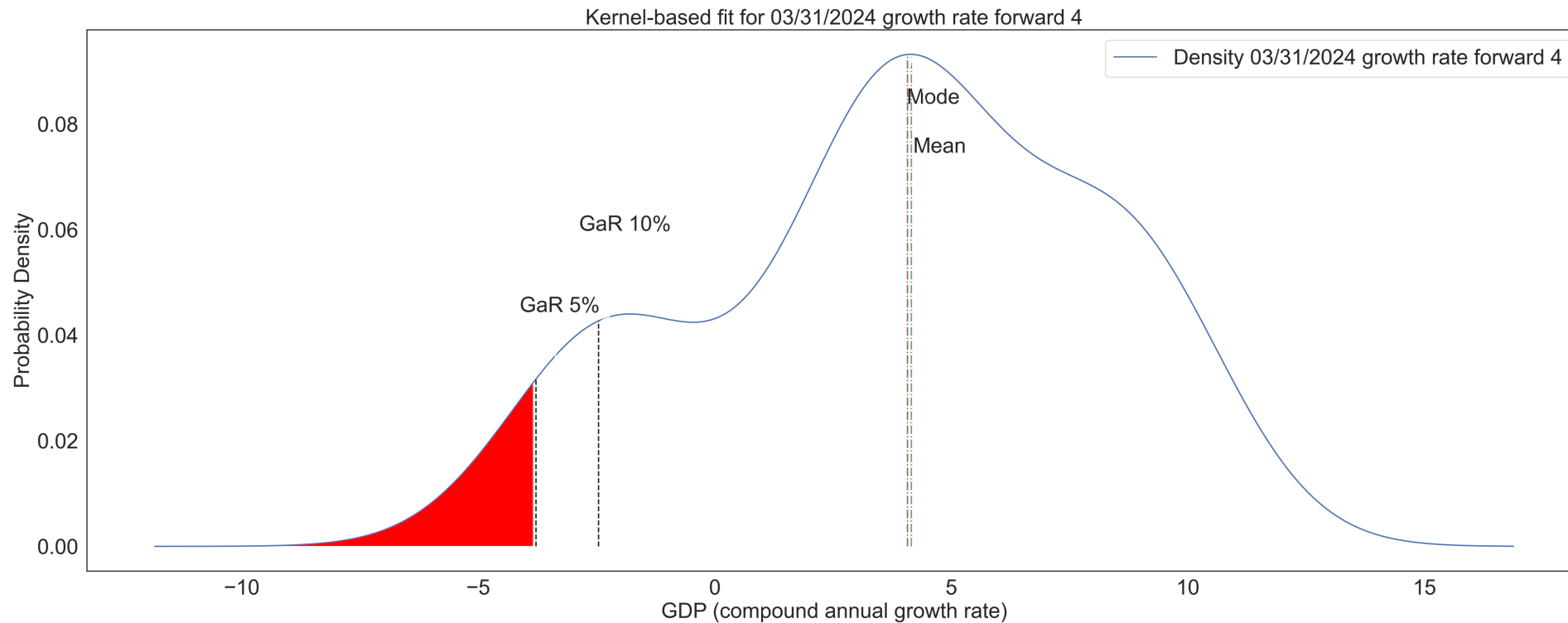
Where is Armenia today?

The distribution for 2024 shows a mode of around 4%. The GaR is -5%. Thusly, the **distance to tail** is 9%. We can also observe that the distribution is left skewed.



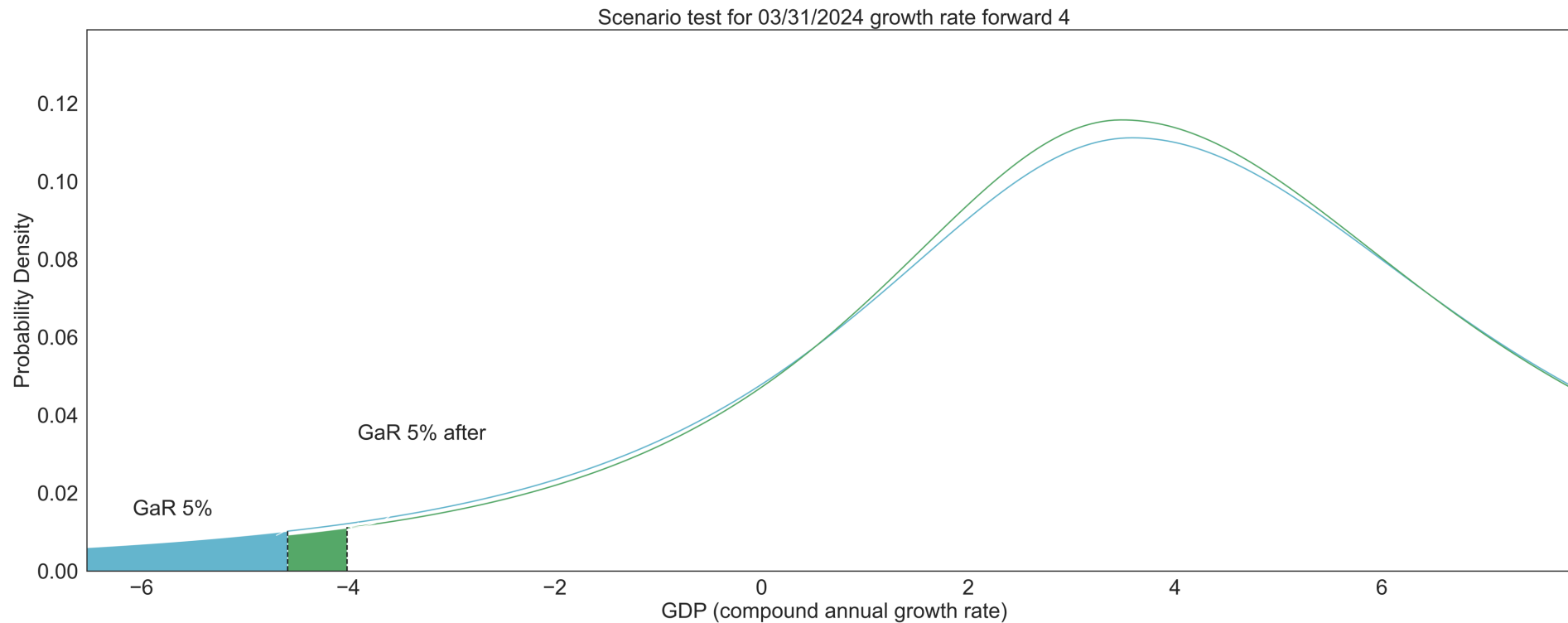
Where is Armenia today?

The **kernel-based estimation** allows us to more closely detect and follow the accumulation of systemic risks stemming from the financial system.



Comparing distributions

Given two arbitrary yet close distributions that offer a **trade-off** in the risk-reward paradigm, what should be our preference?



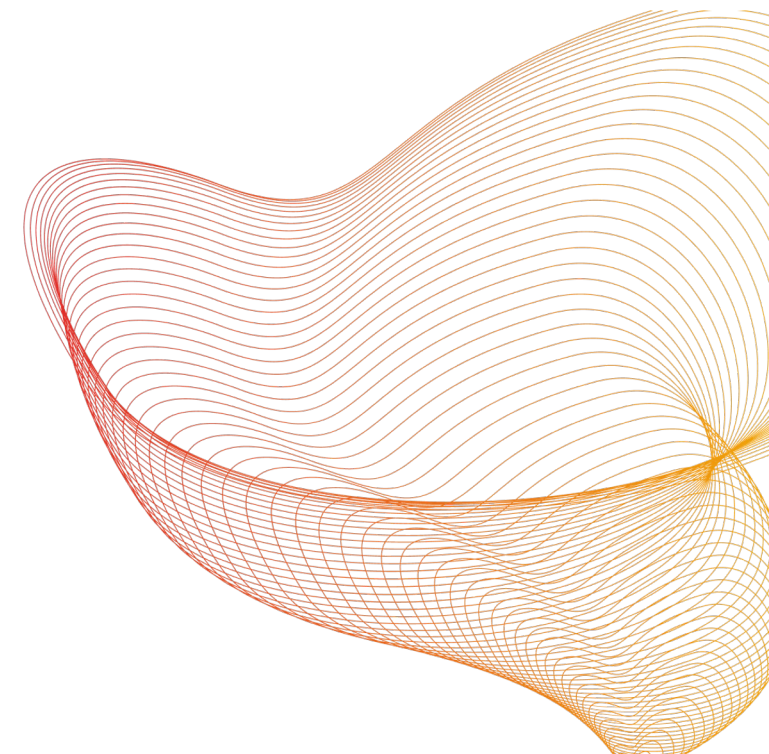
Selecting the better distribution

Welfare function by Suarez (2022):

$$W = \bar{y} - \frac{1}{2}w(\bar{y} - y_c)^2,$$

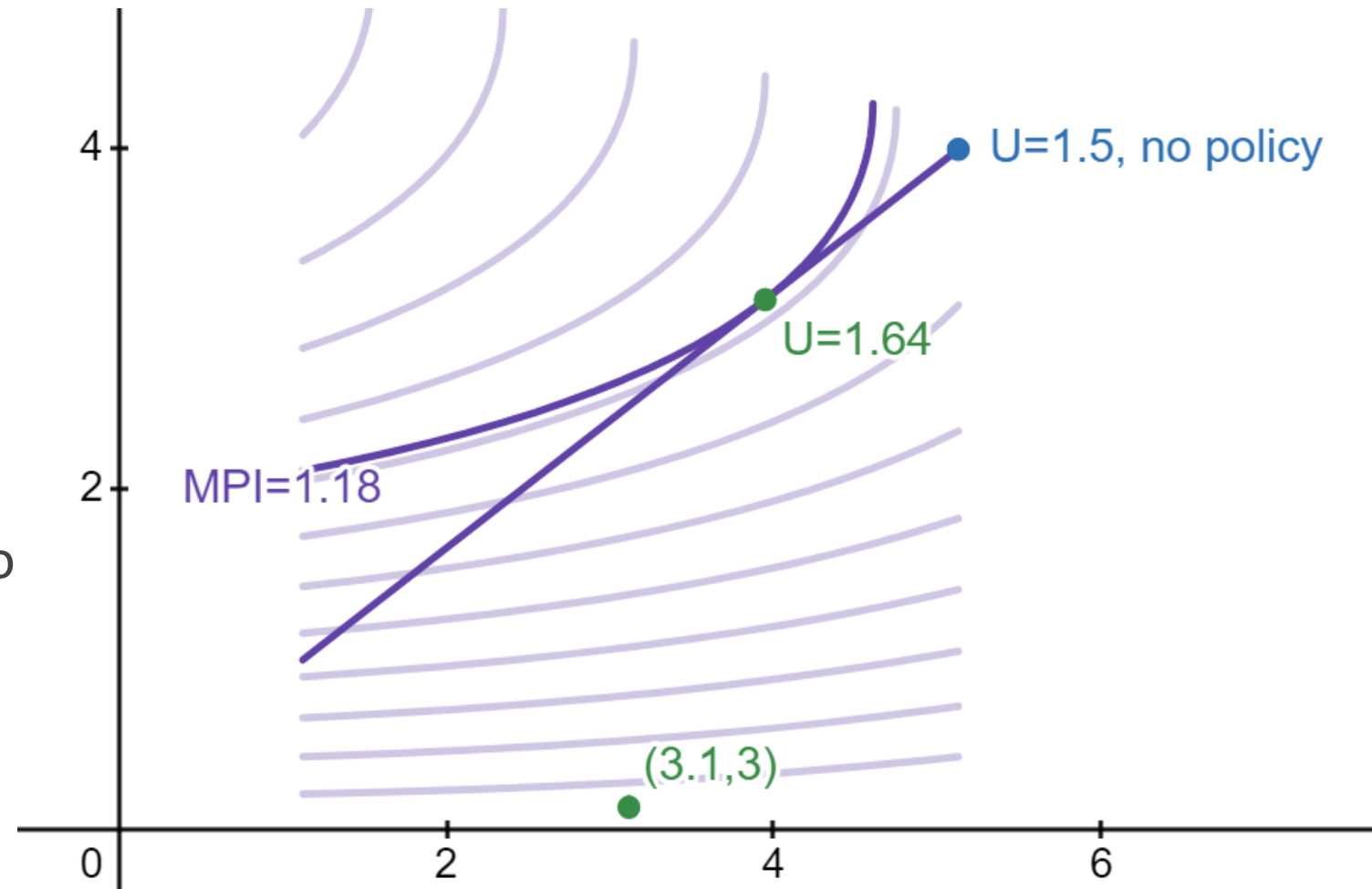
Loss function in general form:

$$\min_{\rho_t} \mathcal{L} = E_t \left\{ \sum_{i=0}^T \beta^i [f(G @ R_{t+i} - \phi y_{t+i})] \right\}$$

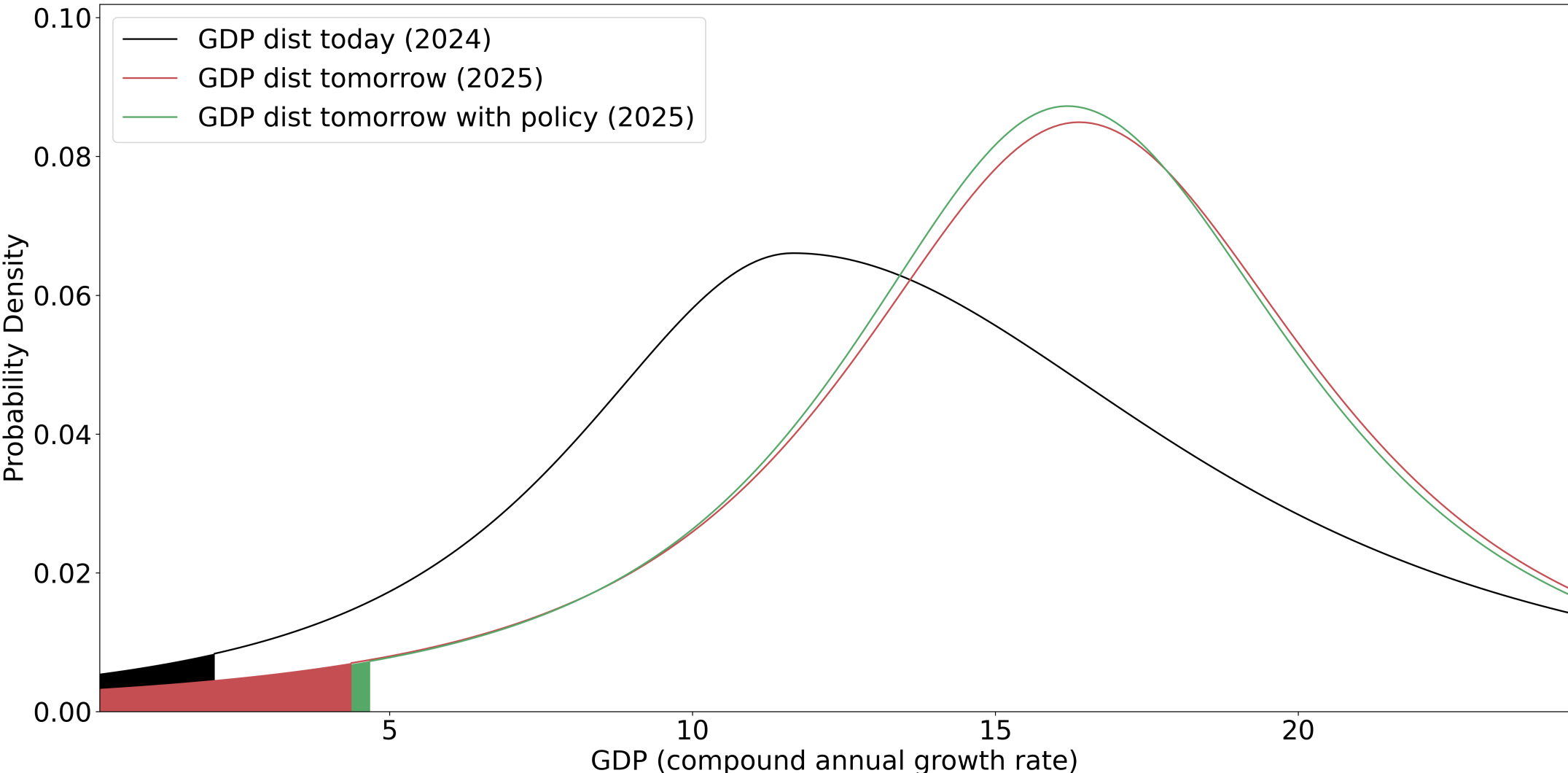


Possibility line, Suarez indifference curves and optimal policy

The **optimal policy** is subject to the choice of “w” term, which we can interpret as the **aversion for financial instability**. This includes the policymakers **appetite for risk** but also the probability of shock.

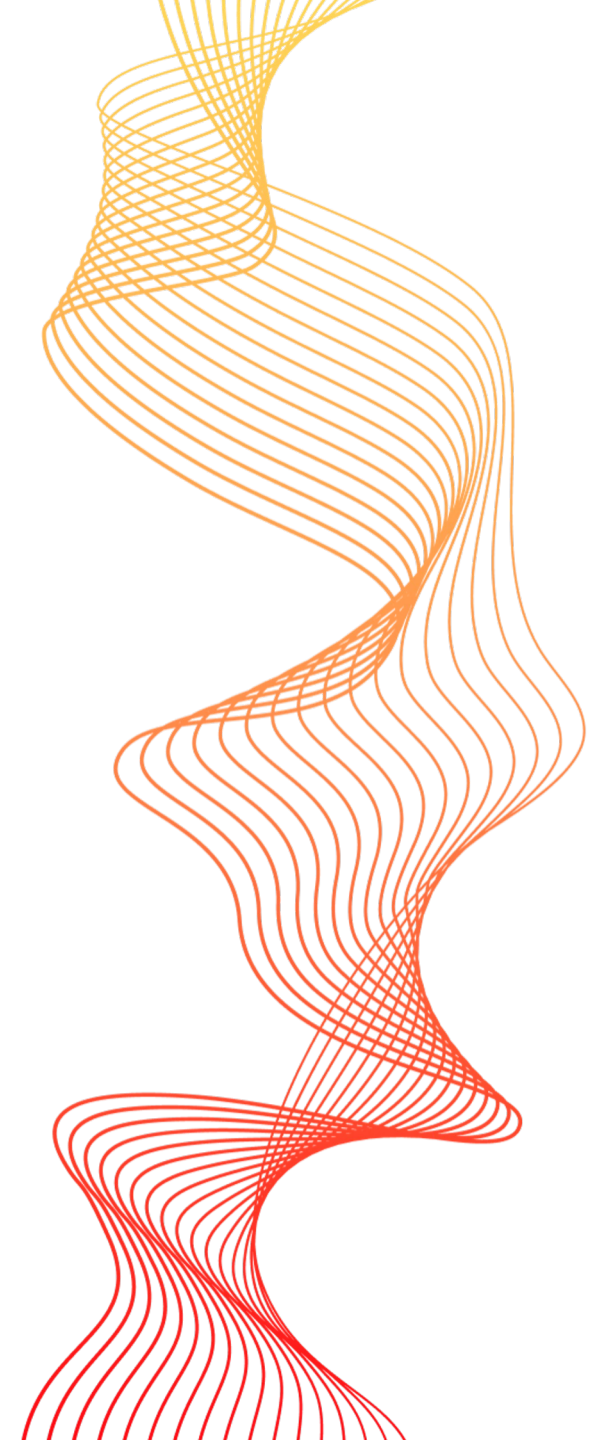


One year ahead GDP distribution forecast and optimal policy calibration



Limitations

1. **Endogeneity** – subject to “reverse causality,” and reduced-form regressions cannot estimate the endogenous changes
2. **Transmission channels** – econometrics methods are unable to model transmission mechanisms, specifically how the MP effects will propagate through the financial system to the real economy.
3. **Limited data**
4. **MPI is an aggregate indicator** – as a result of generalization, the specific effects of the MP instruments are lost. MPI is better suited for determining the direction of MP, but the specific tool or the magnitude of the policy action.

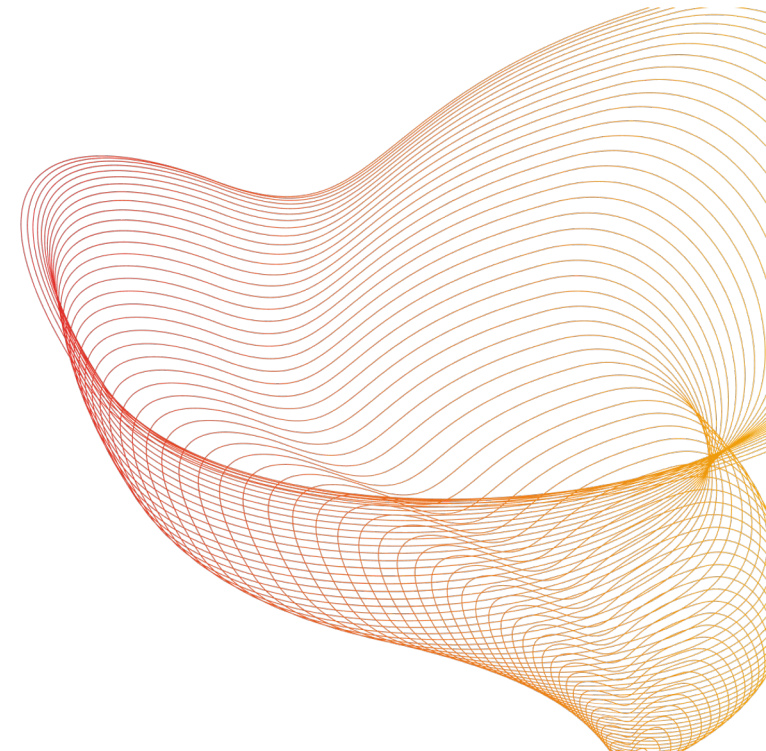


Conclusion

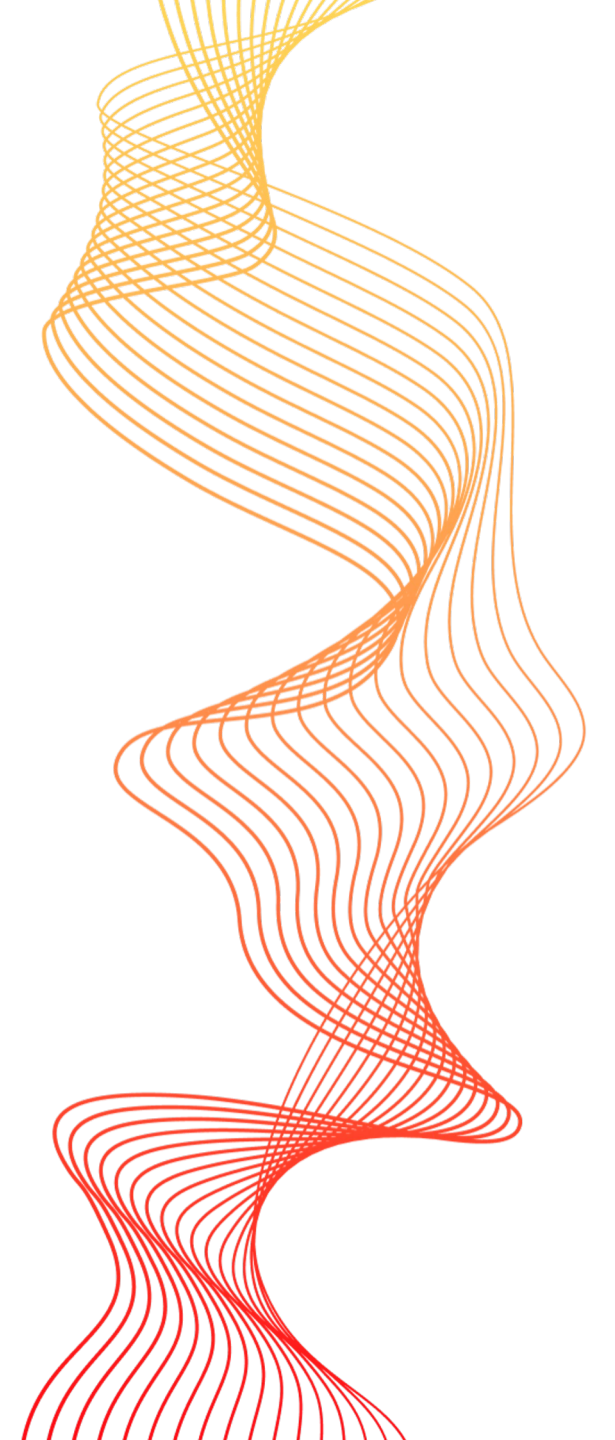
This project aims to quantify systemic risks and the effects of macroprudential policy on them, through the **use of empirical growth-at-risk** (GaR) frameworks.

It aims to augment the process of **decision-making** by proposing methods for designing a policy rule for macroprudential policy.

This rule shouldn't be viewed as a binding constraint, but rather as **guideline** that could serve as a basis for preemptive action.



Thank you

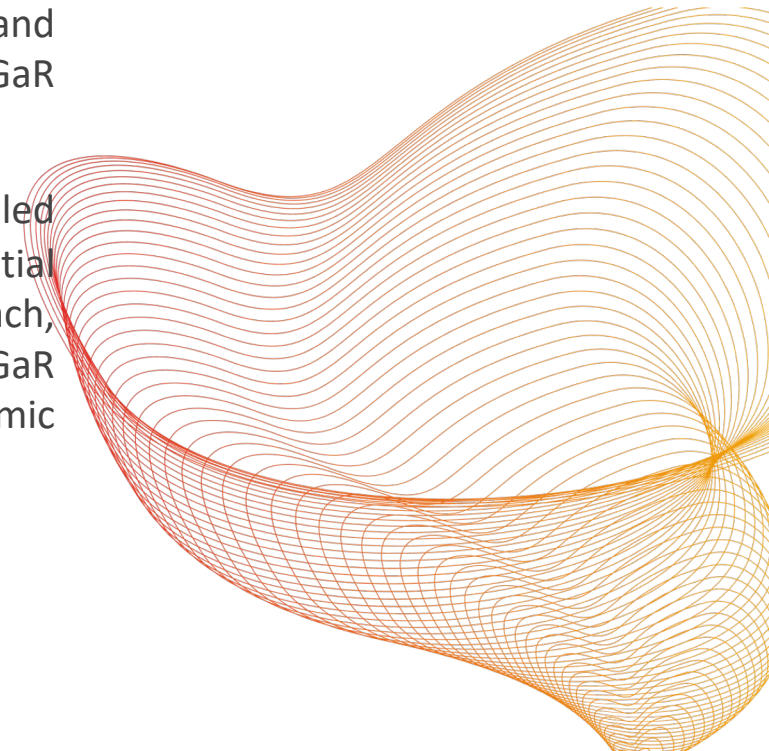


Appendix

Literature review

Several prominent works published in this area:

- **(Aikman, et al., 2019)** applied GaR to a set of 16 advanced economies and found that credit booms and house price overvaluation episodes preceding crisis episodes are expected to significantly deteriorate GaR measure (which is lowest 5th or 10th percentile of GDP growth distribution) in the medium term.
- **(Prasad, et al., 2019)** describe the results of use of GaR by IMF in Peru, Portugal and Singapore, and also present a standardized and easy-to-use Excel based tool for GaR estimation.
- **ESRB (ESRB, 2021)** acknowledges 3 main approaches for measuring the so called “macroprudential stance”, which is essentially the measure of direction of macroprudential policy based on difference between risk accumulation level and resilience: 1. GaR approach, 2. semi-structural models and 3. indicator-based approach. Authors view the GaR (technically, the distance between lower percentile and median) as a measure of systemic risk in the system and examine the impacts of systemic risk and stress parameters on it.



Literature review

Several prominent works published in this area:

- **(Galan, 2020)** conducts analysis based on a panel of 24 advanced and 12 emerging economies and finds, that effects of macroprudential policy can have differentiated impact on GDP mean growth and GDP tail based on number of dimensions (the phase of financial cycle, the direction of macroprudential policy and the type of instrument etc.). His findings confirm, that macroprudential policy effects have more medium-term impact when used in the expansionary phase of the cycle, while their use in downturns have more immediate and smaller effect.
- **(Suarez, 2022)** offers a simple, yet very effective way for macroprudential policy decision rule by designing a welfare function, that connects future expected GDP growth with its and a parameter gauging the risk appetite of a policymaker. With his basic theoretical framework he suggests, that policymakers should not target the decrease of the tail of GDP distribution, but the distance between median and tail.

